

Iran Renewable System Analysis

Group A | February 5th, 2026

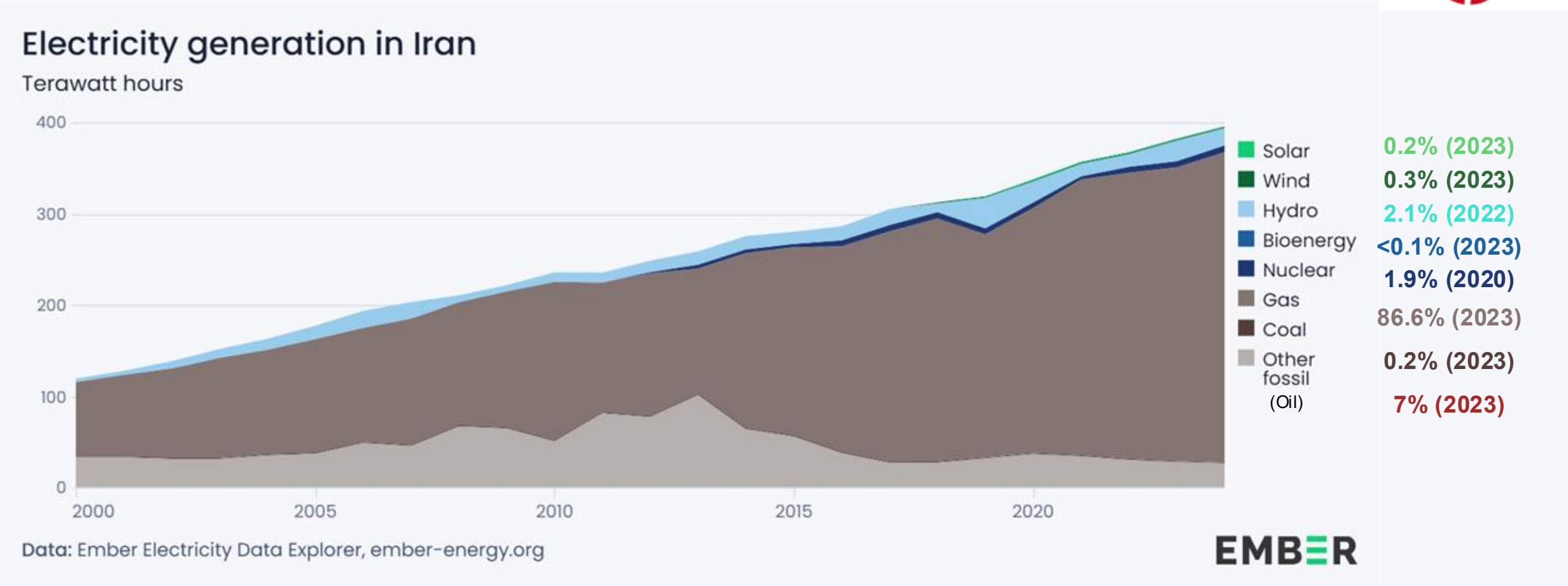
Data Science for Energy System Modelling

Outline



- 1** Introduction
- 2** Model Inputs
- 3** Results
- 4** Sensitivity Analysis
- 5** Limitations

Context - Current State of Electricity System in Iran



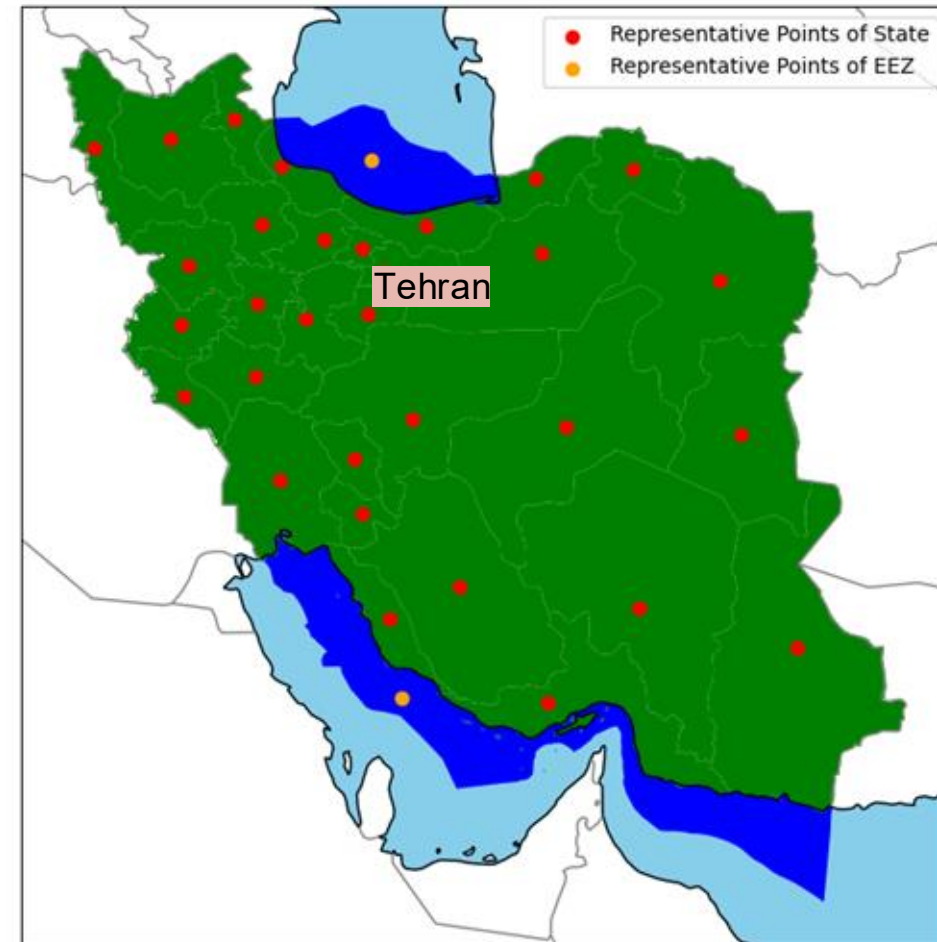
Fact Sources: (IR Gas Production, 2024), (IR Oil Production, 2023)

Context - Model Considerations



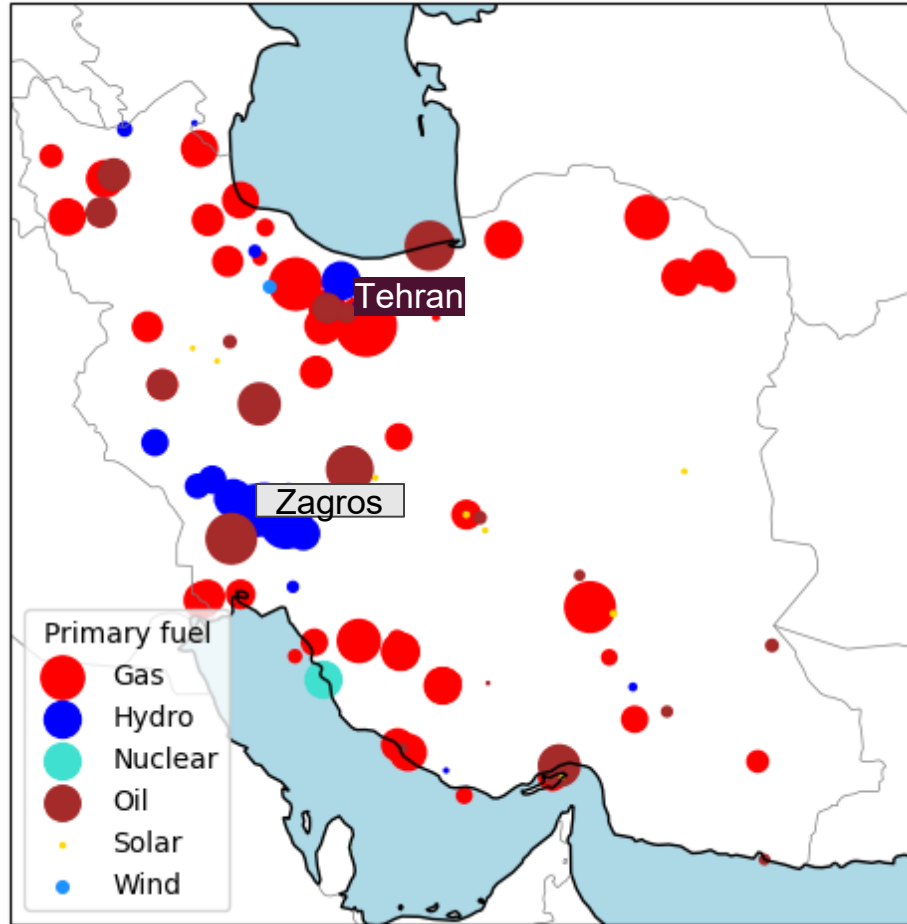
- Area: 1.648 million km²
(vs. Germany: 0.357 million km²)
- Population (2024): ~85,961,000
(vs. Germany ~83,600,000)
- System costs sourced from various currency years
- Cost projection year: 2025
- Discount Rate: 7%
- 31 States + 2 exclusive economic zones (EEZ)
- 1 Bus per state/EEZ
- Buses set at regional representative point

Sources: (Bundesamt, 2024), (Statistical Center of Iran, 2024)

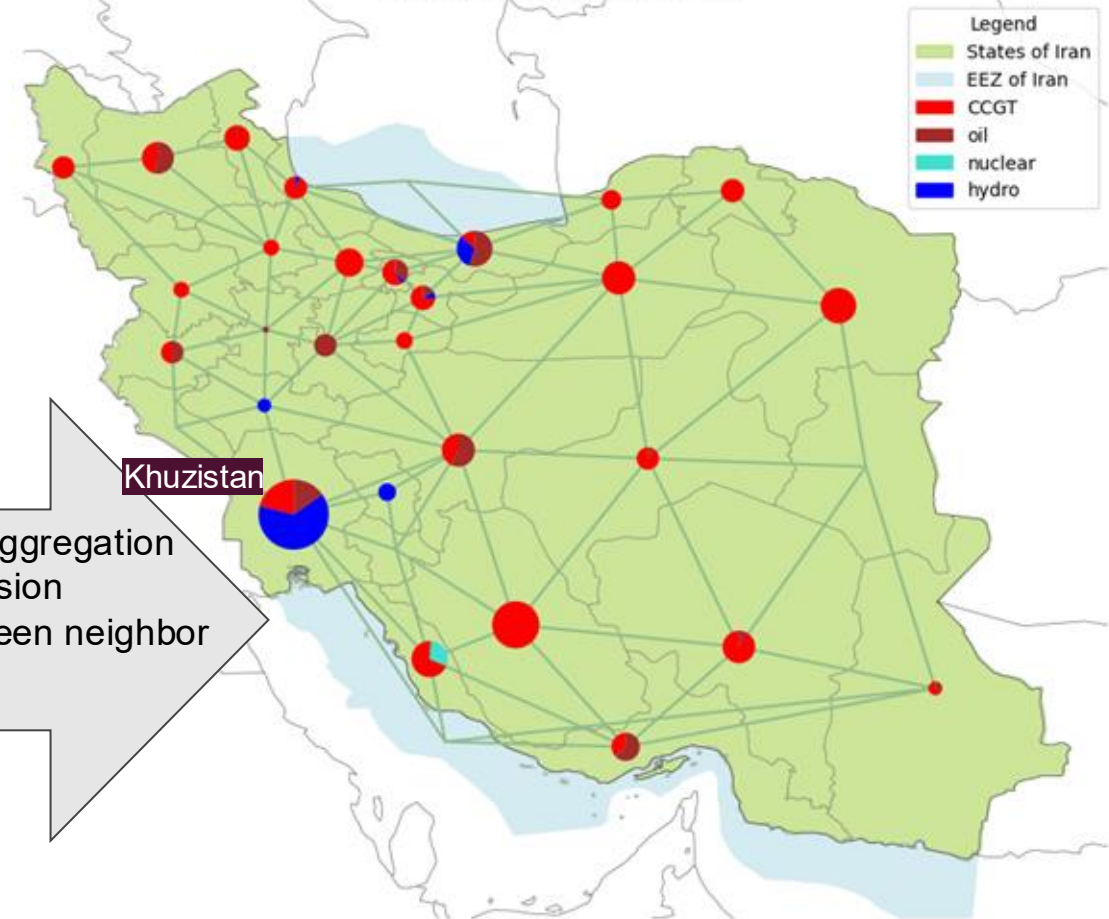


Inputs - Power Plant Capacities

Location of power plants and their primary fuel

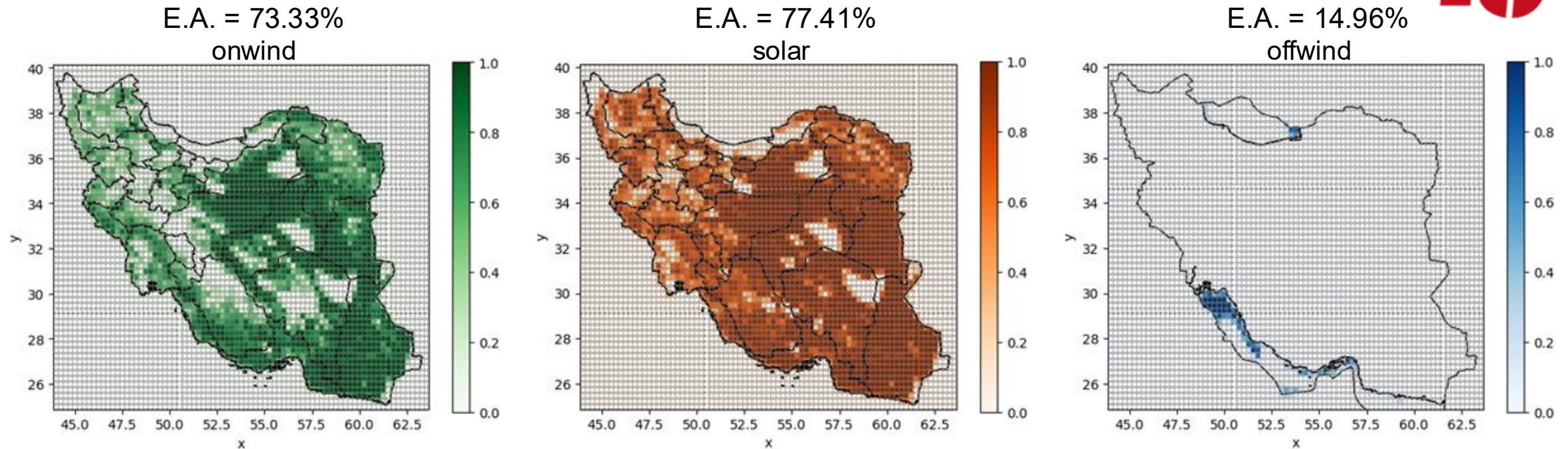


Existing Power Plants per Region



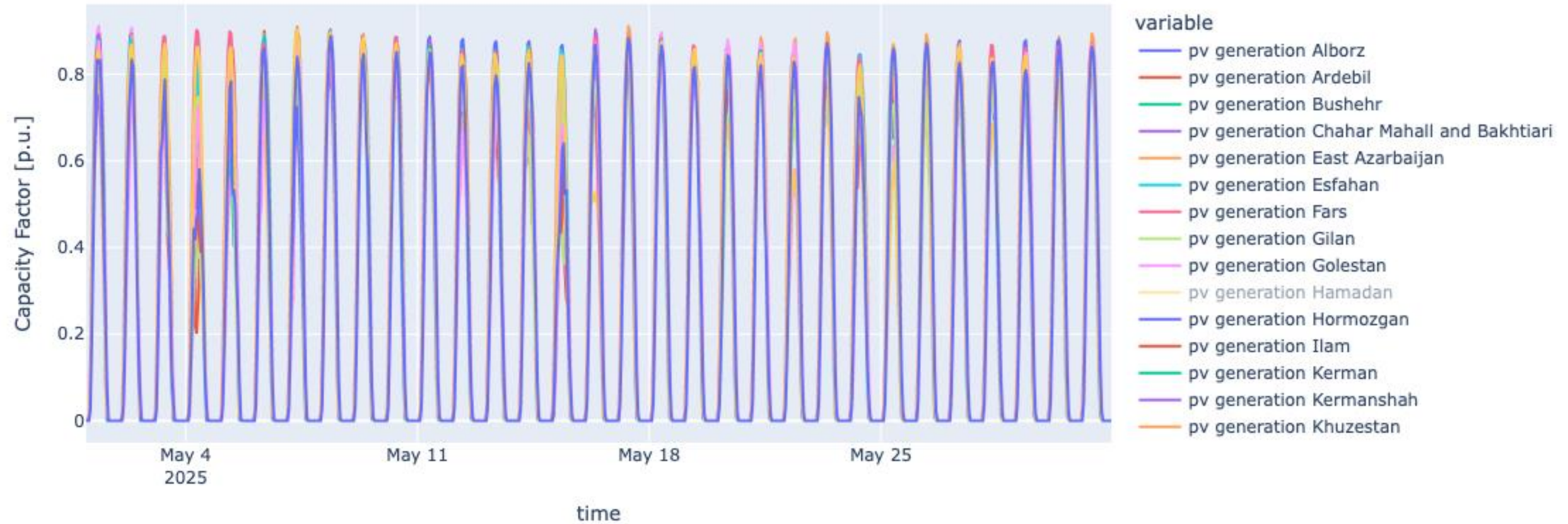
- Regional Aggregation
- RES Exclusion
- Links between neighbor regions

Inputs - Land Eligibility Analysis

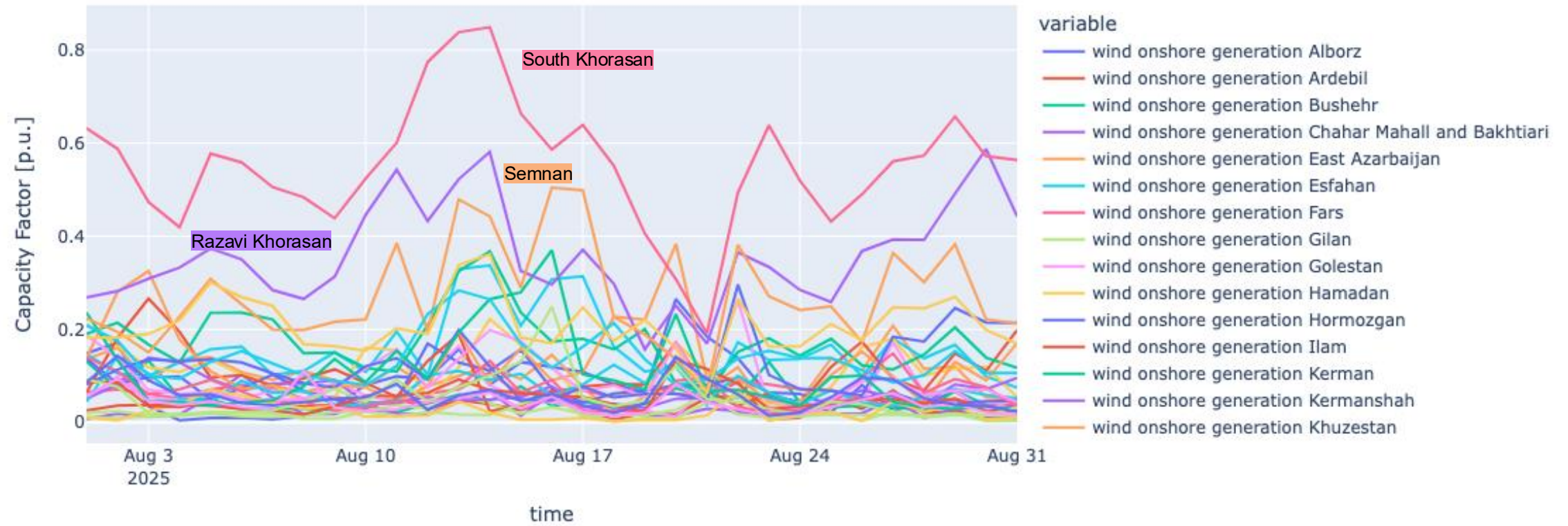


- Considered criteria:
 - Elevation data
 - Distance from airports, major roads, built up areas
 - Exclusion of world protected areas
 - Exclusive Economic Zones (for wind offshore)
 - Land classification

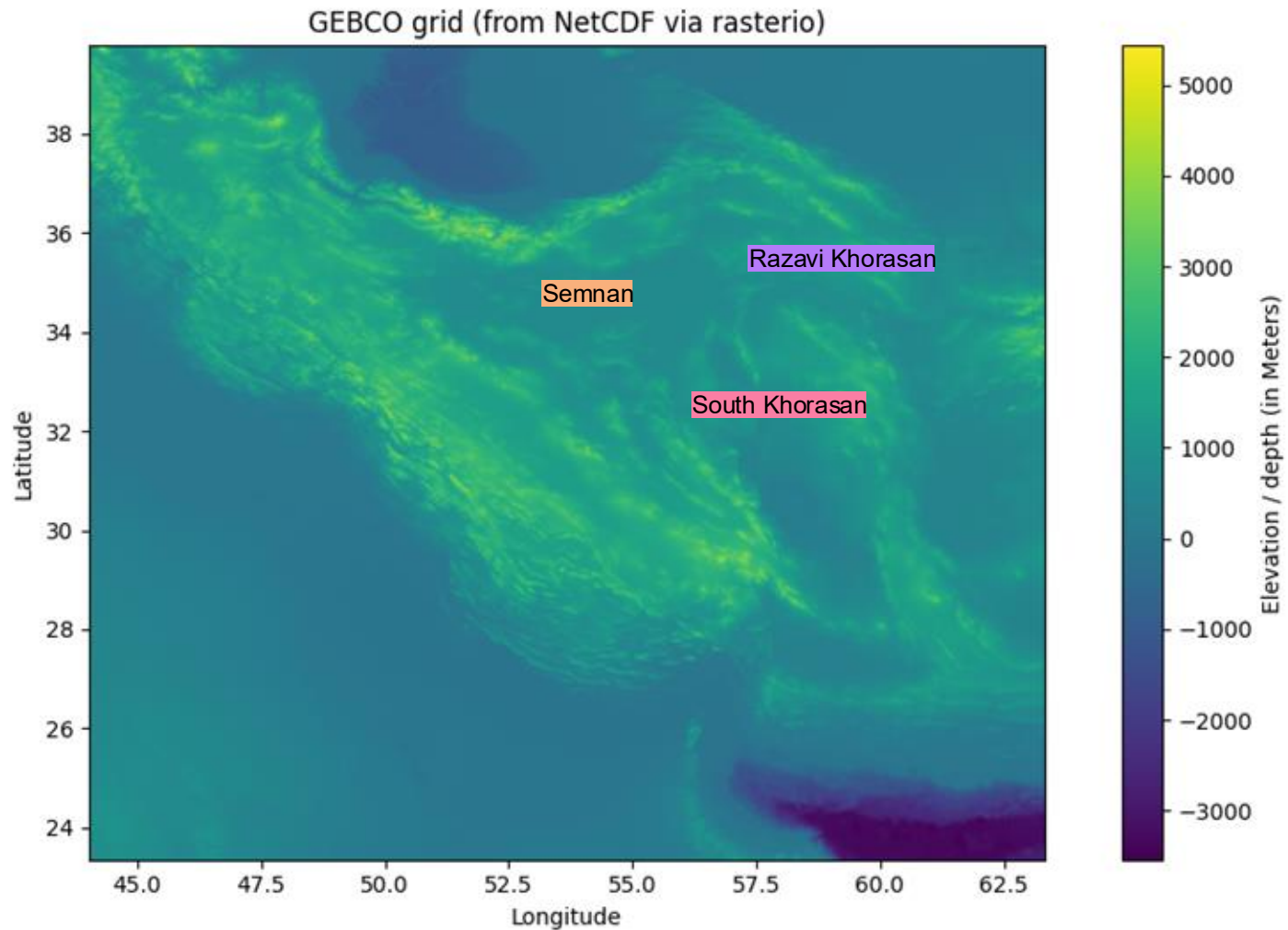
Inputs - RES Time Series (PV - May - 2025)



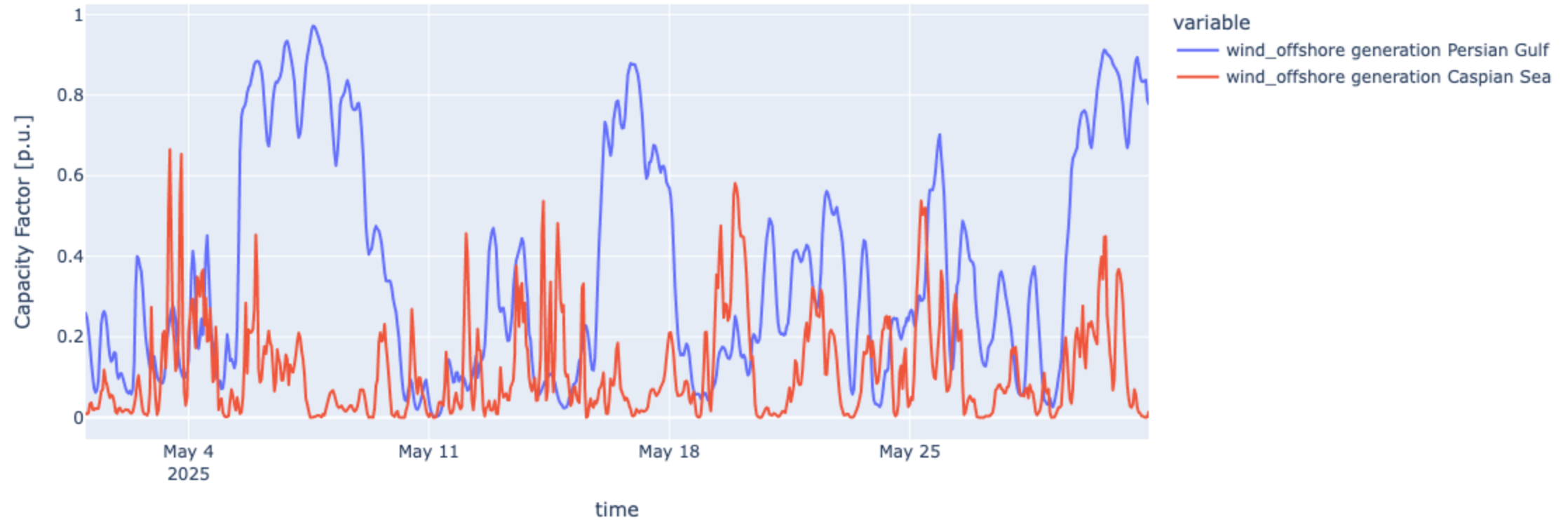
Inputs - RES Time Series (Wind onshore - August 2025)



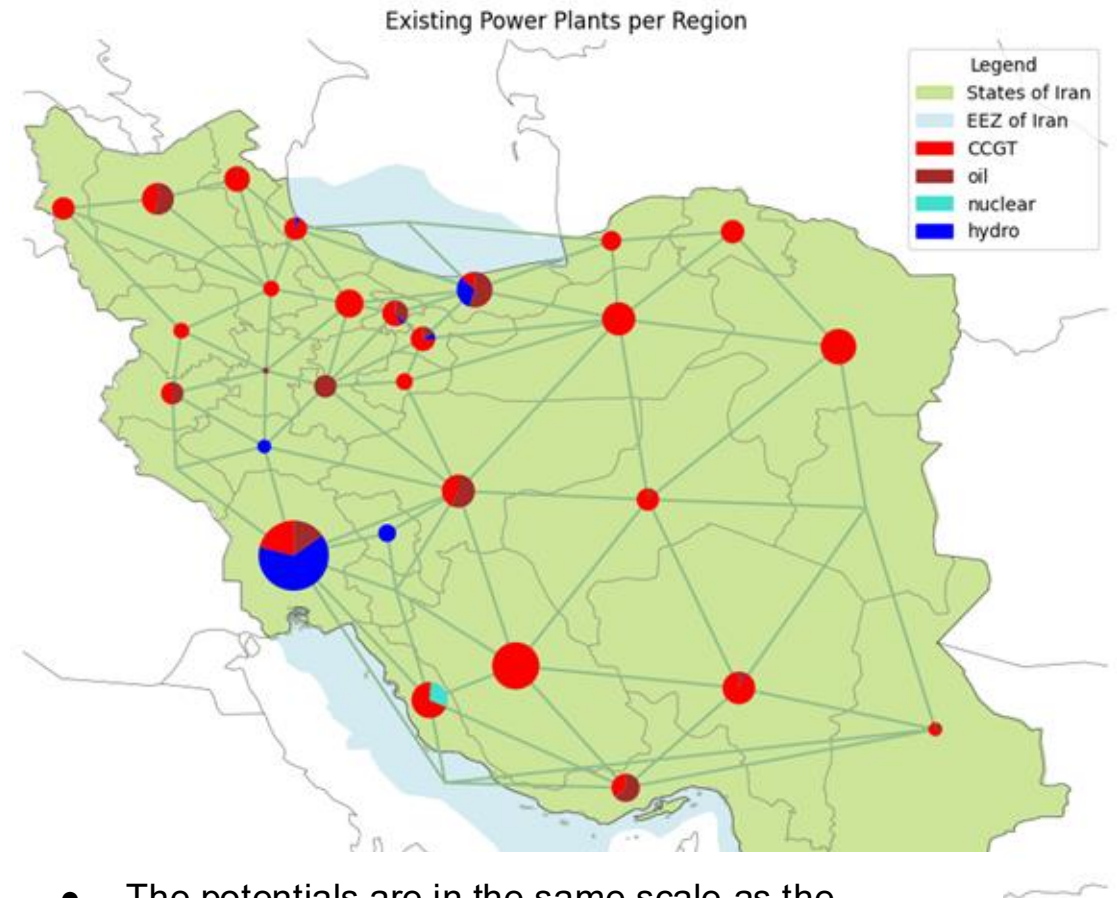
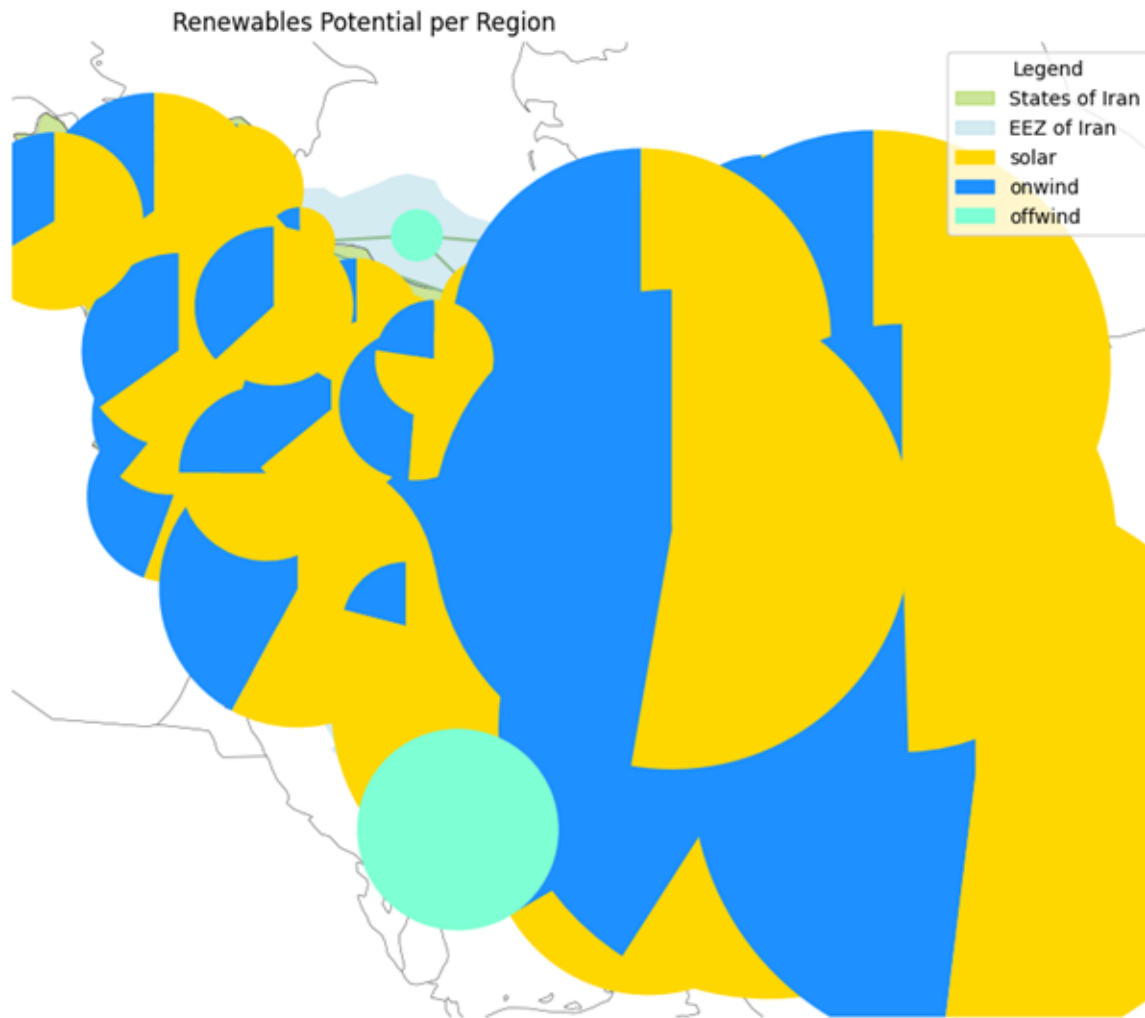
Inputs - Elevation Data



Inputs - RES Time Series (Wind offshore - May 2025)

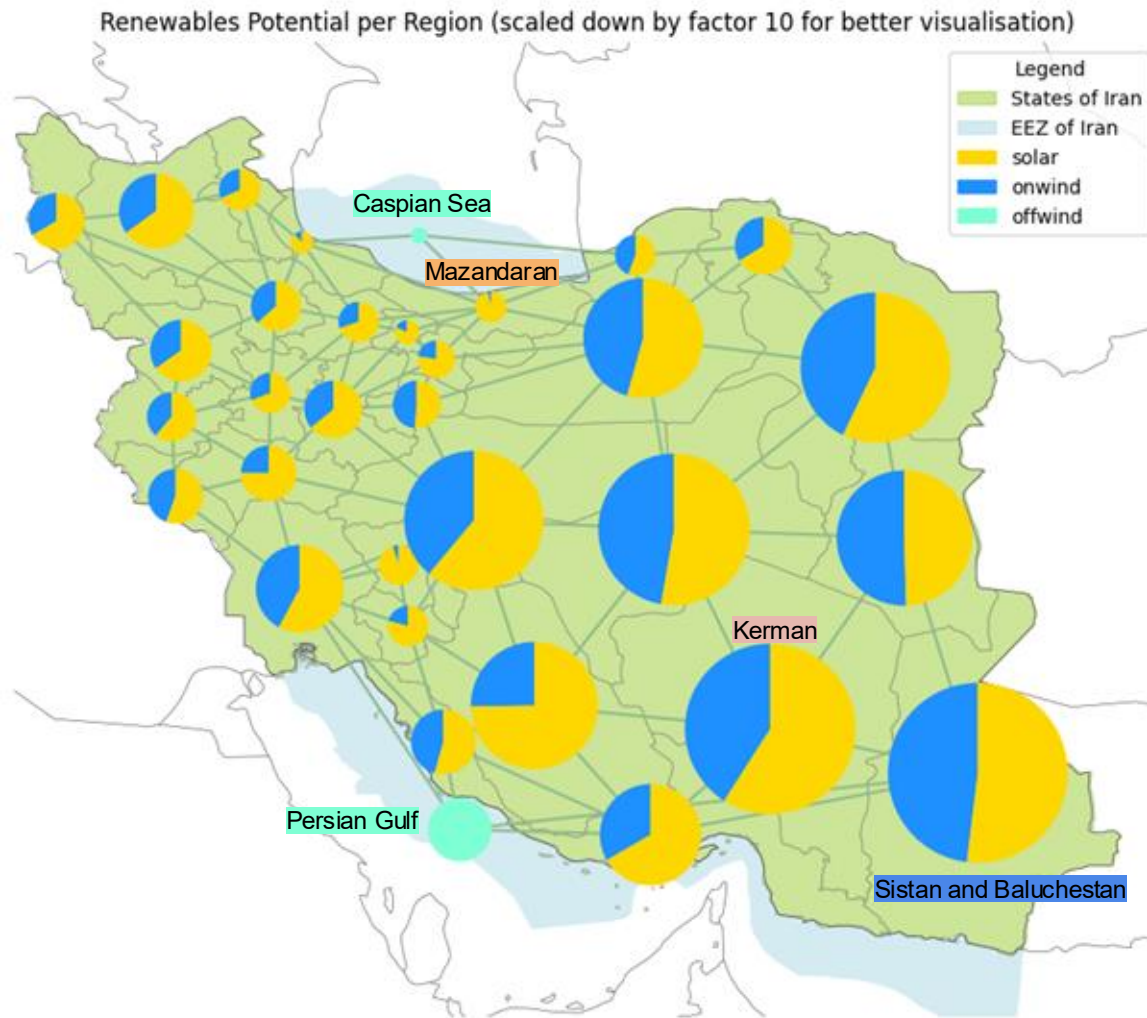


Inputs - RES Available Capacity (same scale of existing power plants)



- The potentials are in the same scale as the previous plot, showing existing power plants!

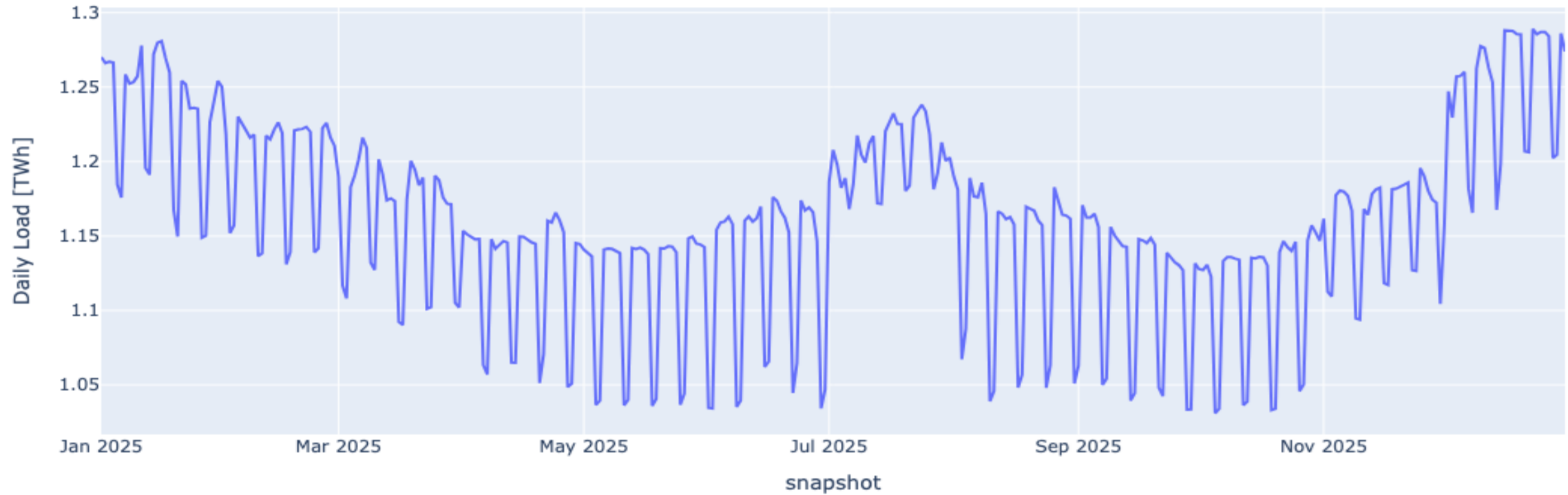
Inputs - RES Available Capacity (scaled down by factor 10)



- Base: Land Eligibility Analysis and 2025 weather
- High Potential of Solar and Wind Capacities, especially in the center and east

- **Sistan and Baluchestan:**
 - Solar: ~380 GW
 - Wind onshore: ~350 GW
- **Kerman:**
 - Solar: ~390 GW
 - Wind onshore: ~270 GW
- **Mazandaran:**
 - Solar: ~21 GW
 - Wind onshore: ~0.8 GW
- **Persian Gulf:**
 - Wind offshore: ~92 GW
- **Caspian Sea:**
 - Wind offshore: ~6 GW

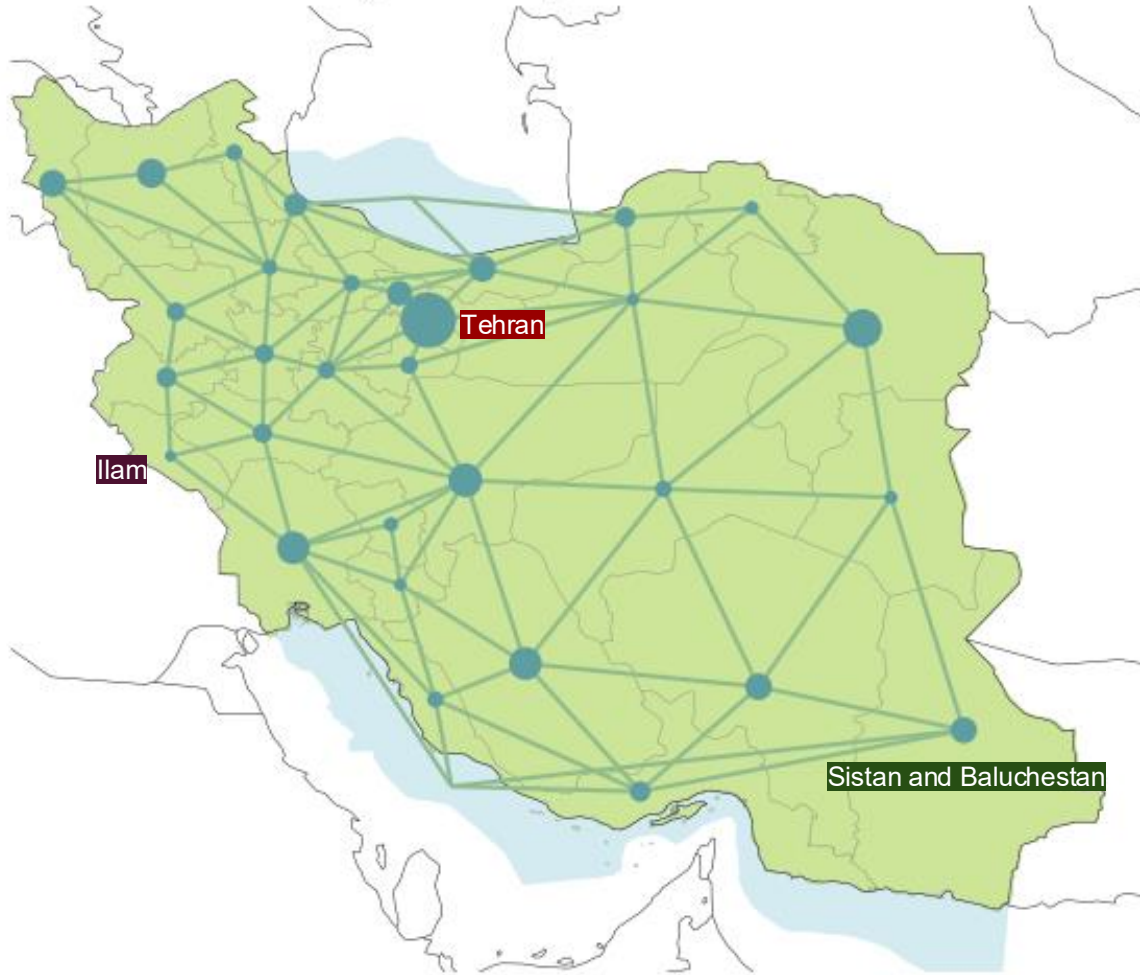
Inputs - Load Time Series



Inputs - Regional Load Distribution

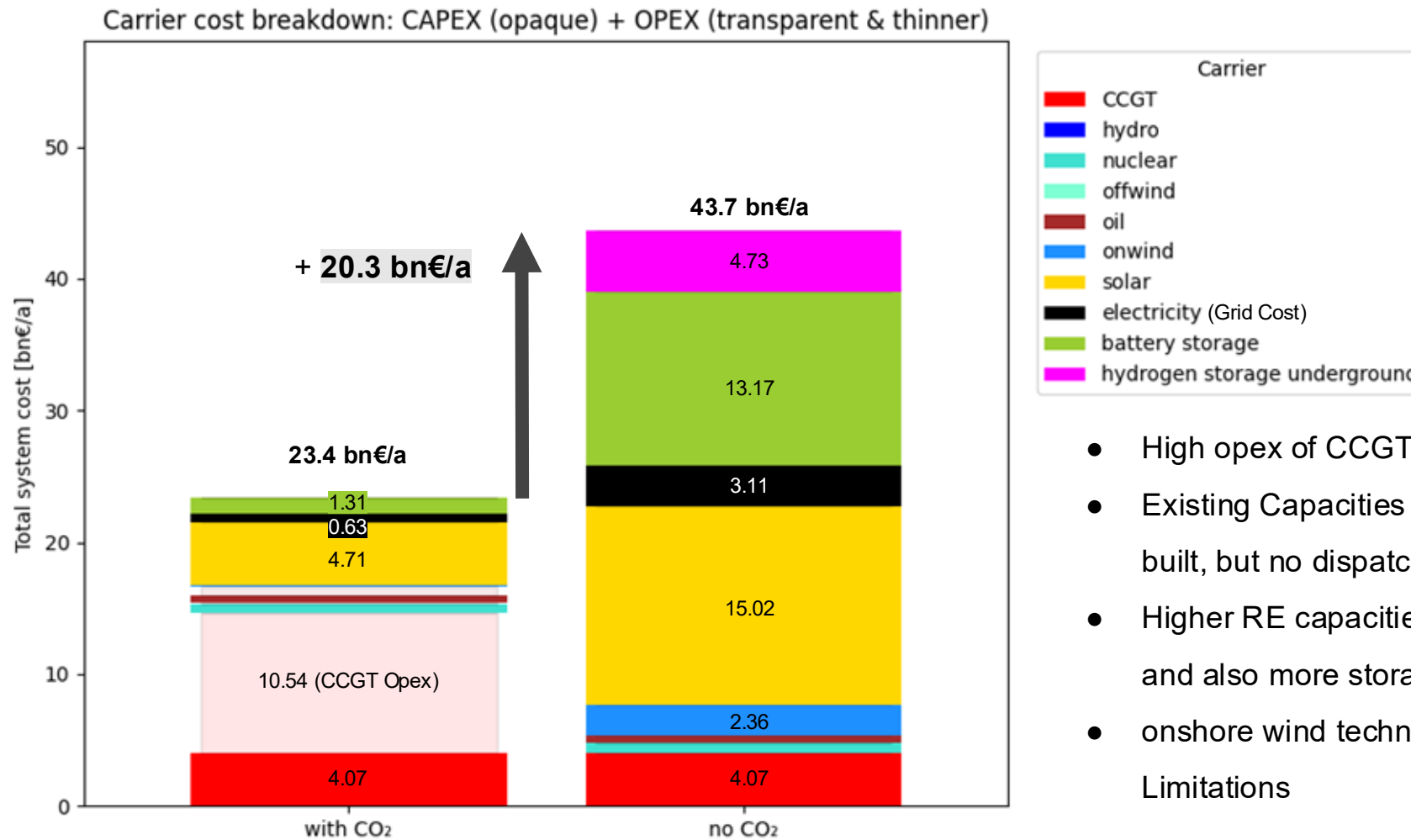


Yearly Electricity Load Distribution



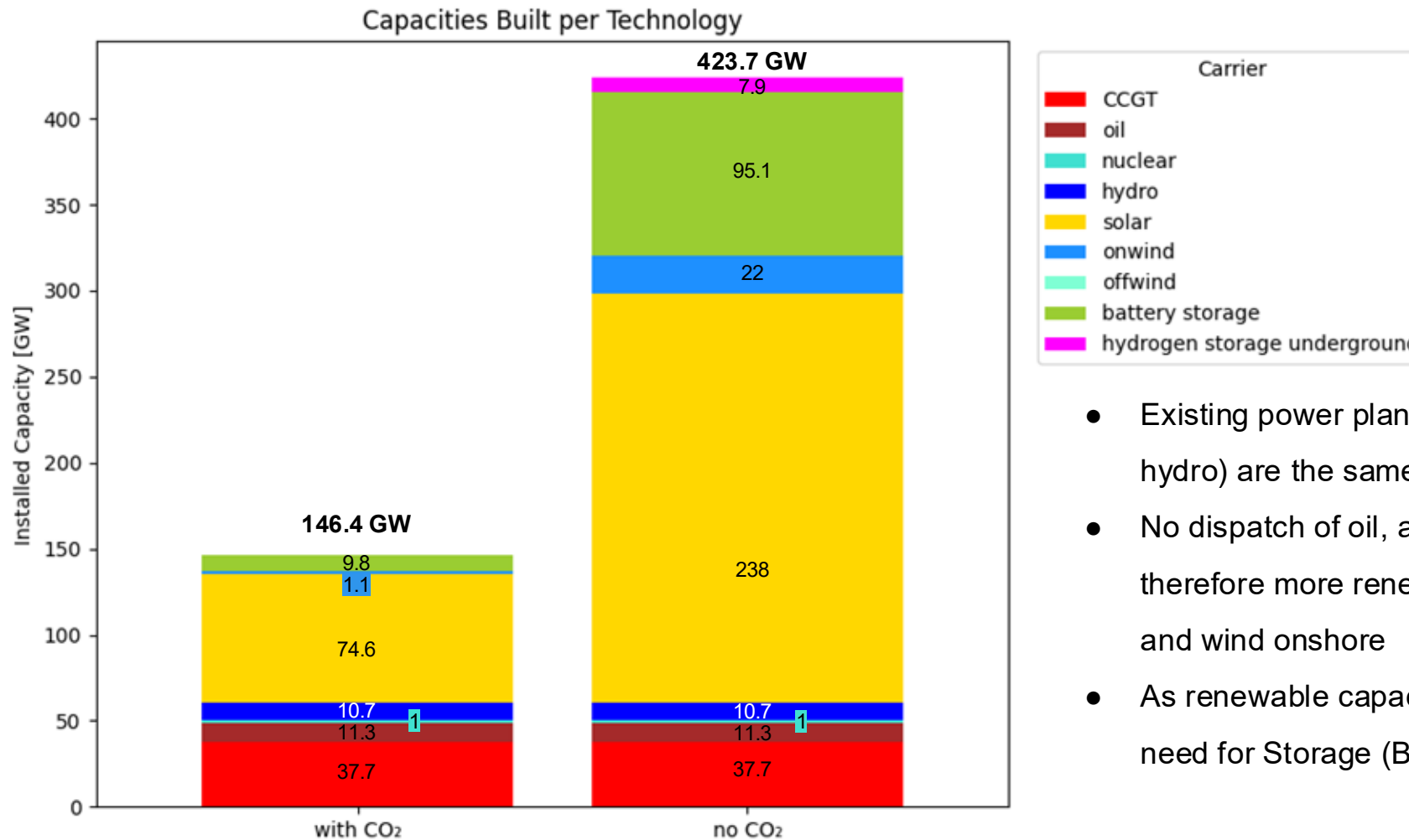
- Load time series (from 2013 projected to 2030)
- Electricity demand of ~424 TWh/a
- Assumption: Distributed to regions, based on population
- Load per Region ranges between:
 - **Tehran:** ~71 TWh/a
 - ...
 - **Sistan and Baluchestan:** ~16 TWh/a
 - ...
 - **Ilam:** ~3 TWh/a

Results - Total System Costs



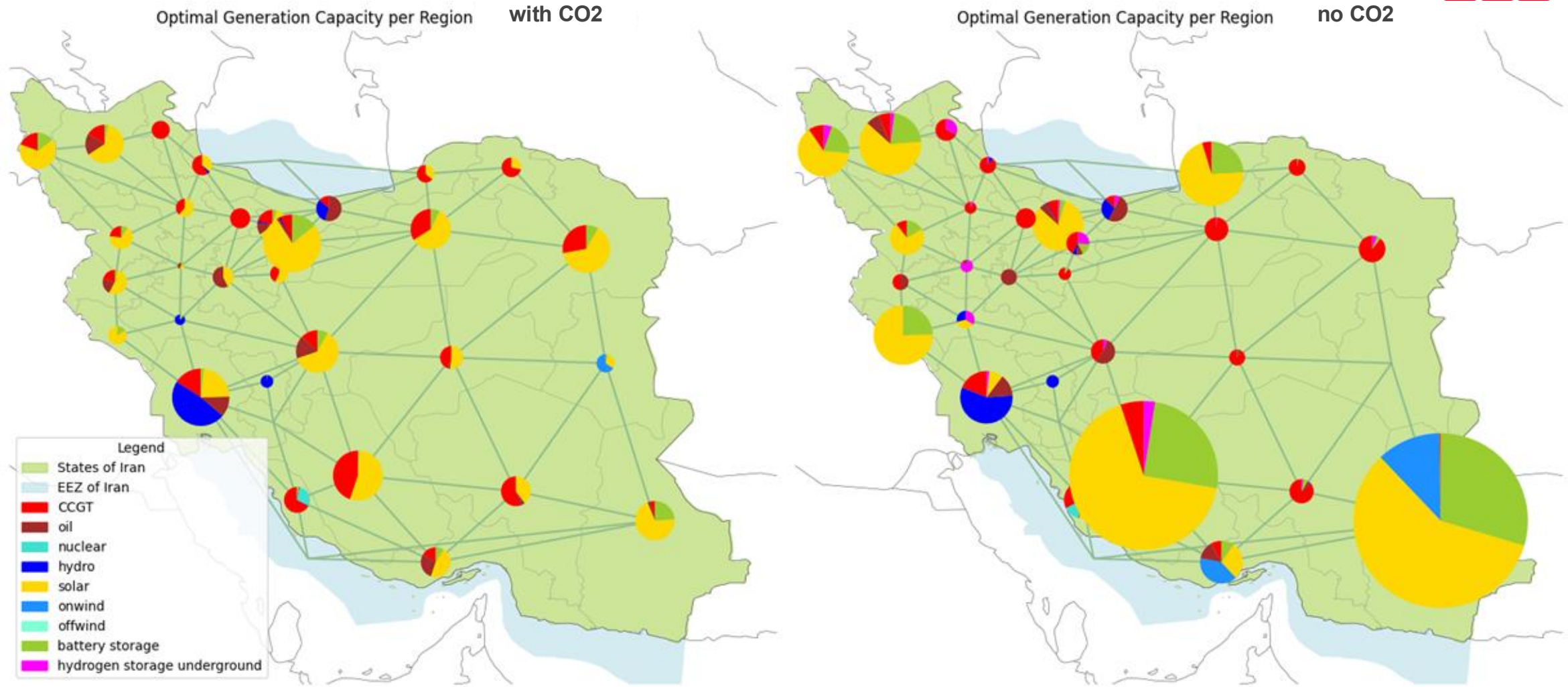
- High opex of CCGT in base case (with CO₂)
 - Existing Capacities (CCGT, Oil, Nuclear) always built, but no dispatch of CCGT and Oil in “No CO₂”
 - Higher RE capacities require more grid investment, and also more storages
 - onshore wind technology only required with CO₂
- Limitations

Results - Optimal Generation Capacity



- Existing power plants (CCGT, oil, nuclear and hydro) are the same in both cases
- No dispatch of oil, and CCGT in “no CO₂” case, therefore more renewable capacities, mostly solar and wind onshore
- As renewable capacity increases, there is more need for Storage (Battery and Hydrogen)

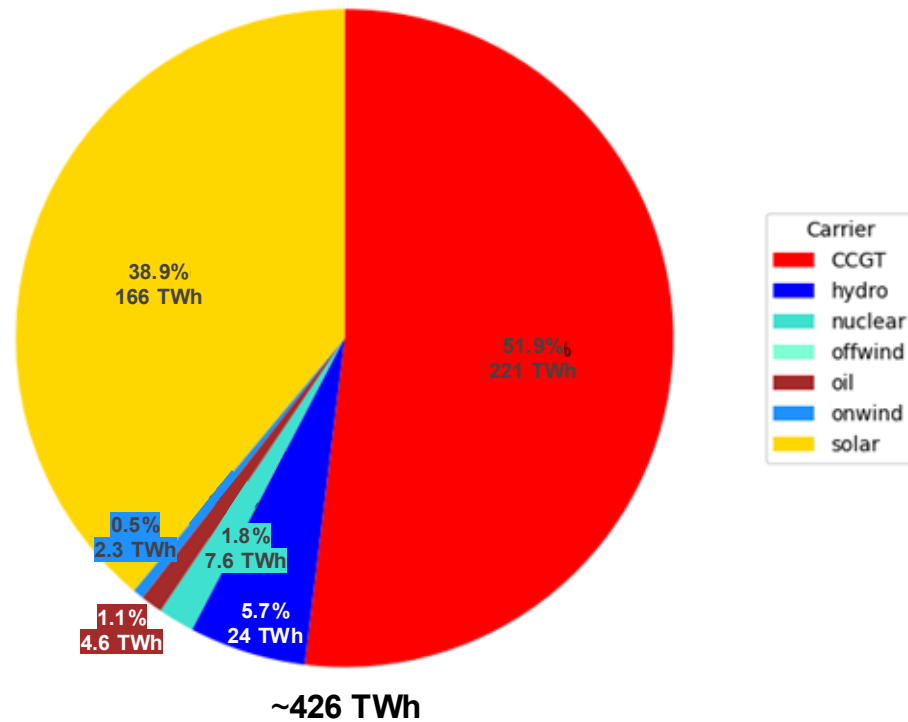
Results - Optimal Generation Capacity per Region



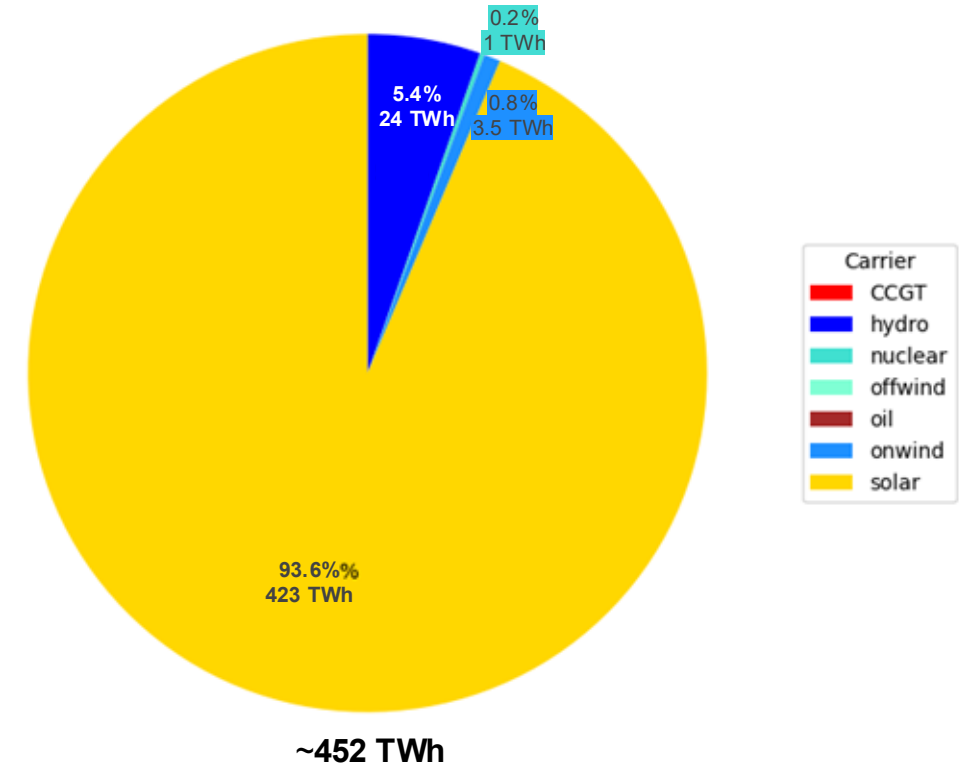
Results - Electricity Mix



Electricity Generation by Carrier (with CO2)

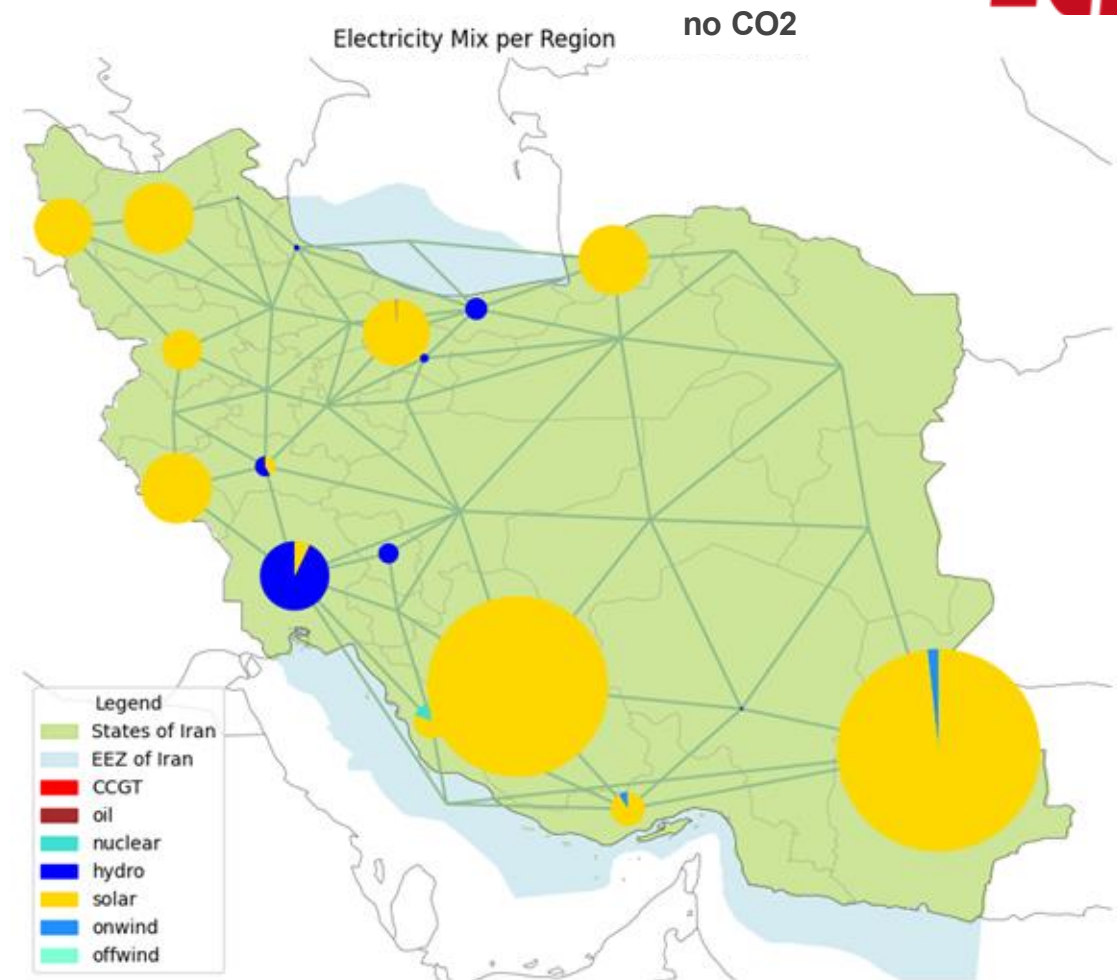
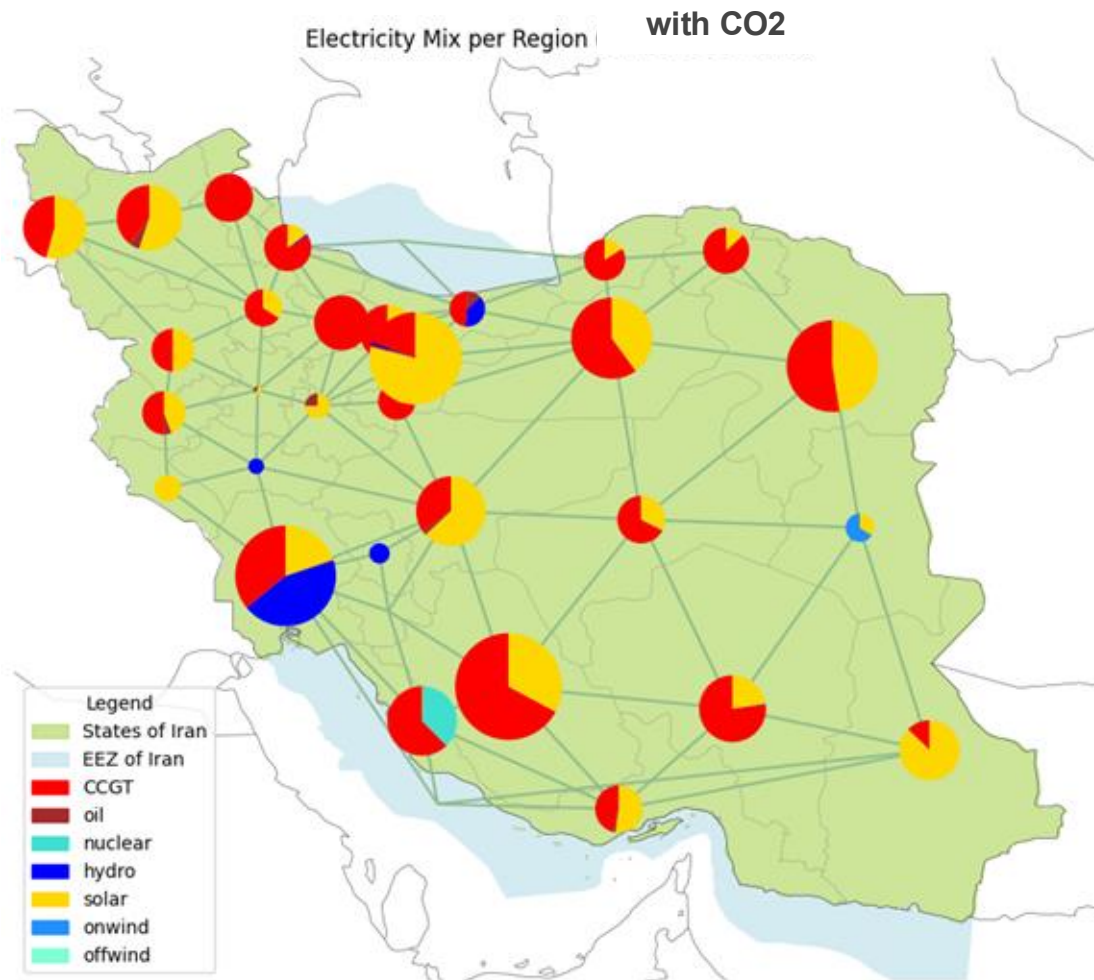


Electricity Generation by Carrier (no CO2)

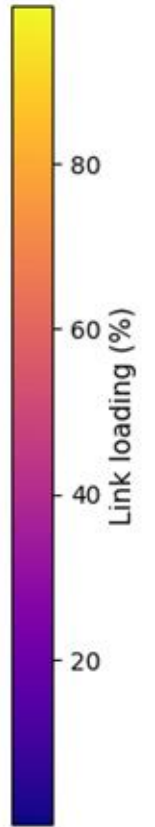
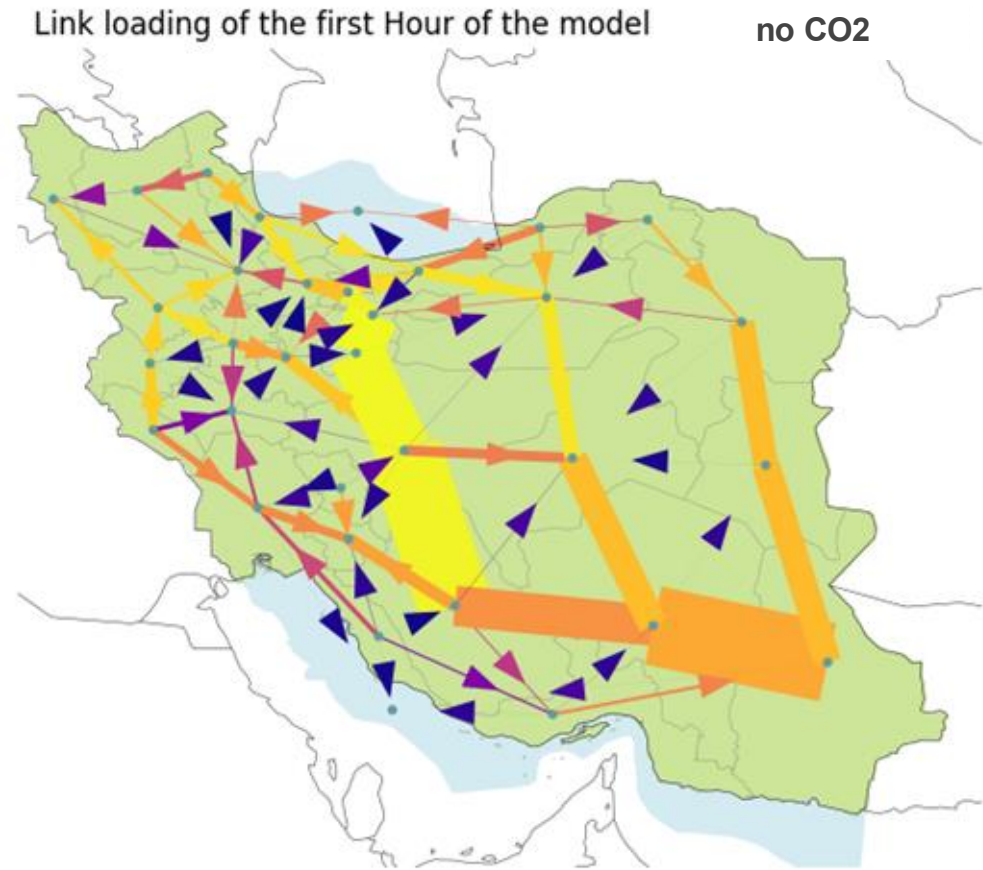
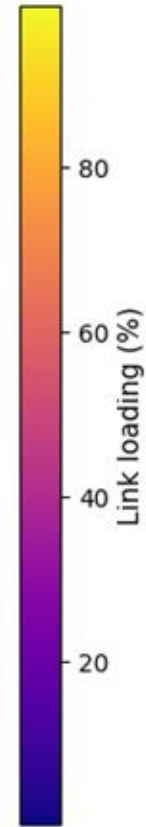
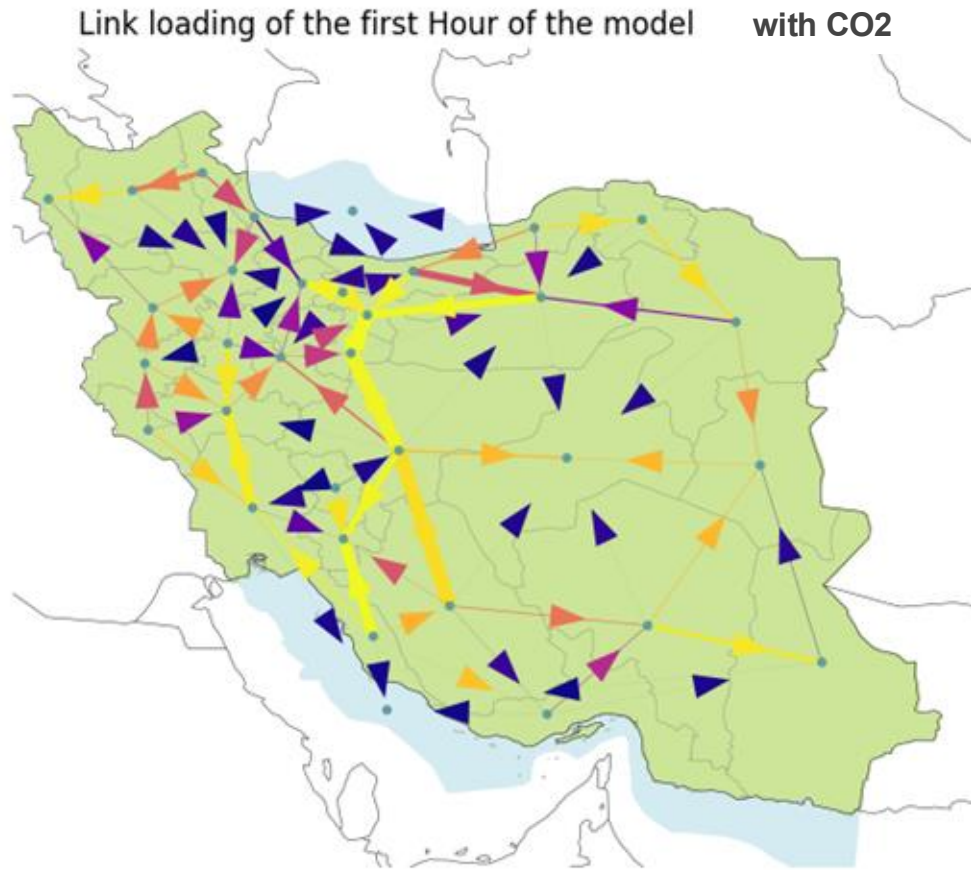


- More generation than load mostly because of the losses in Storages

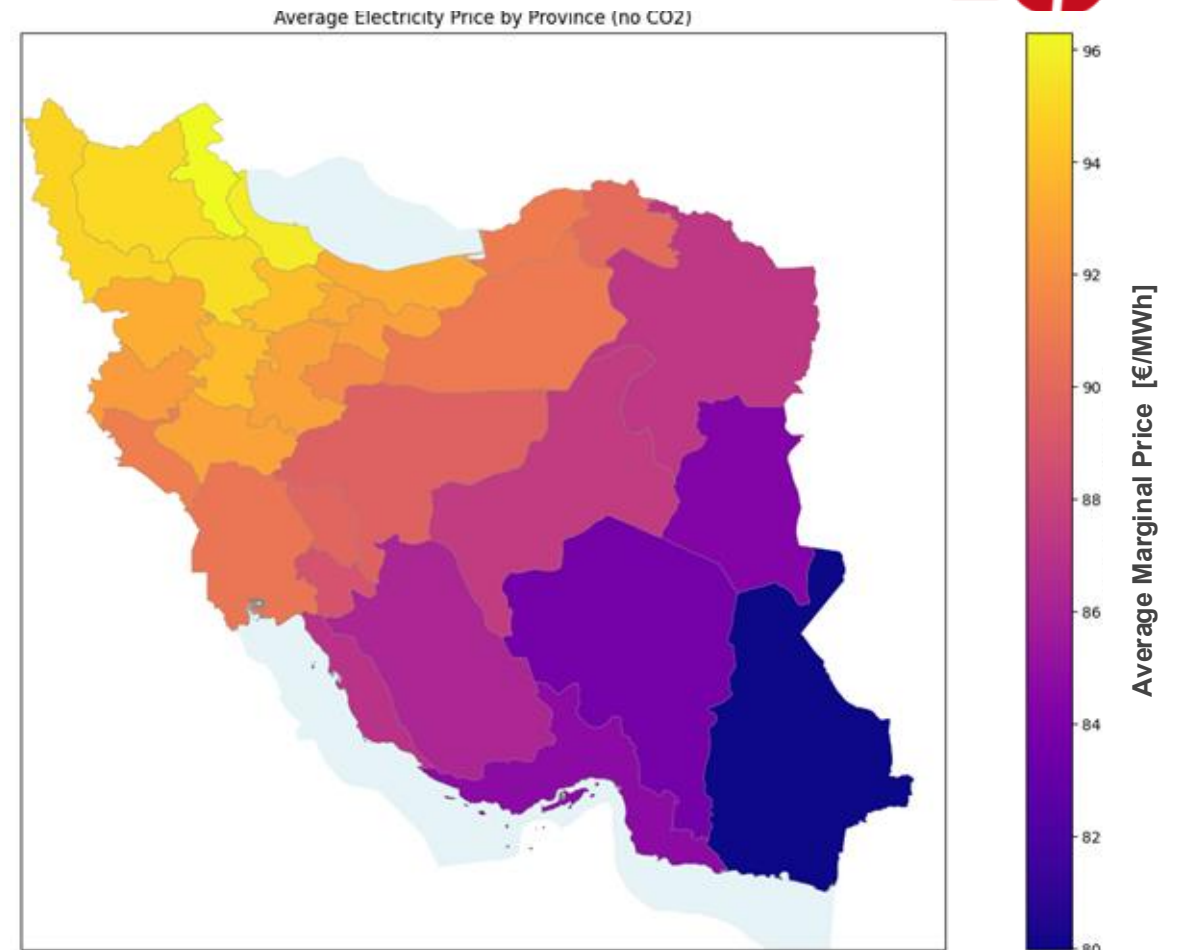
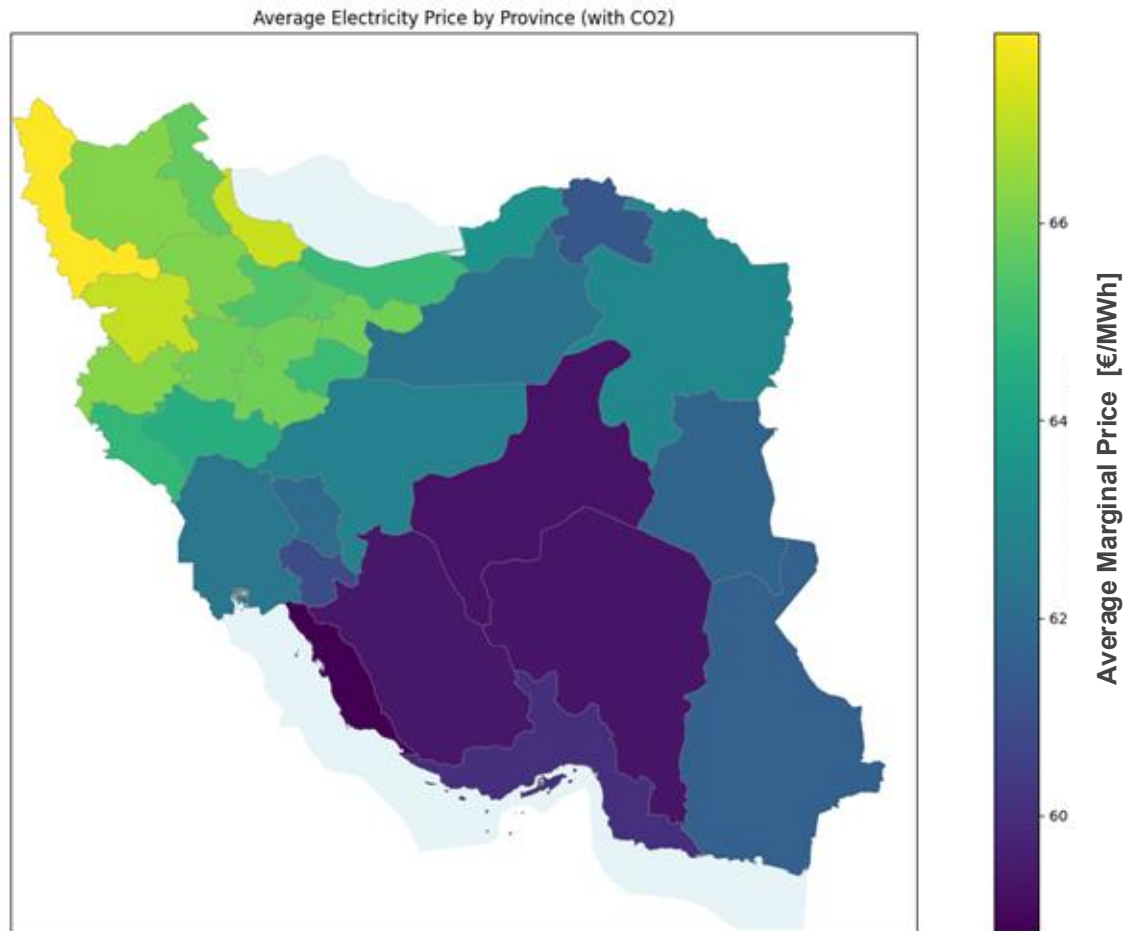
Results - Electricity Generation Mix per Region



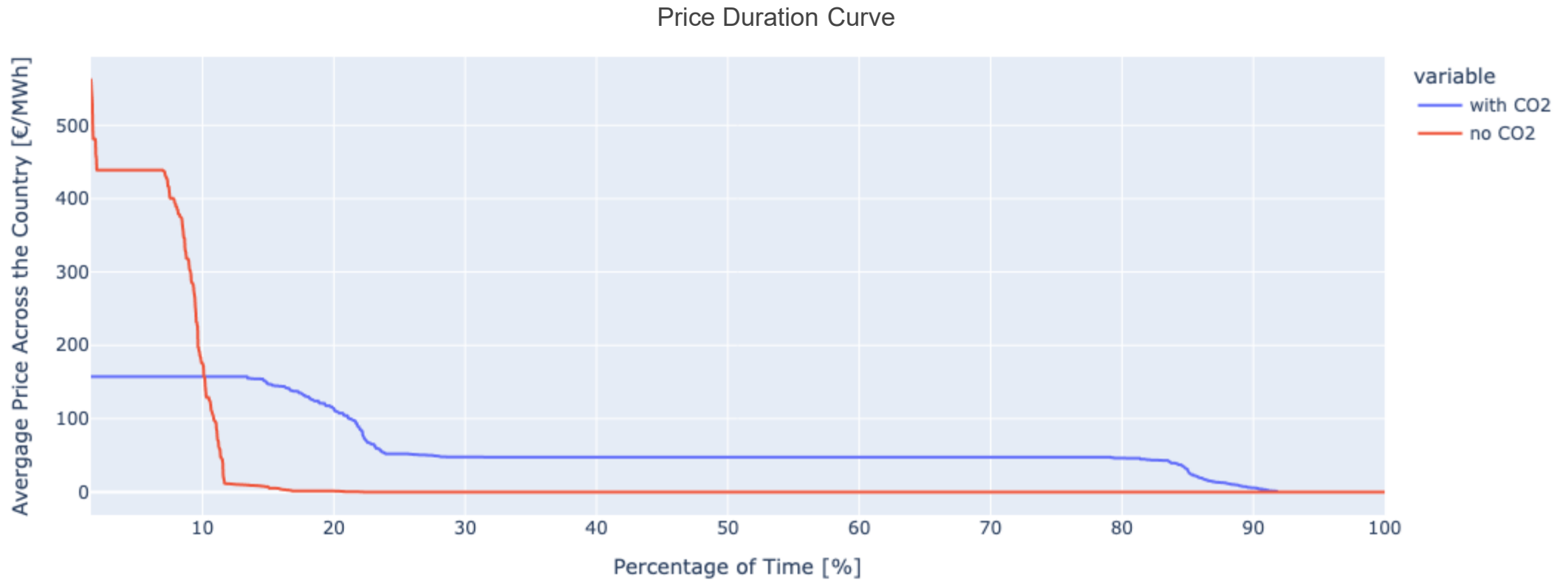
Results - Power Flow



Results - Average Price per Region



Results - Prices

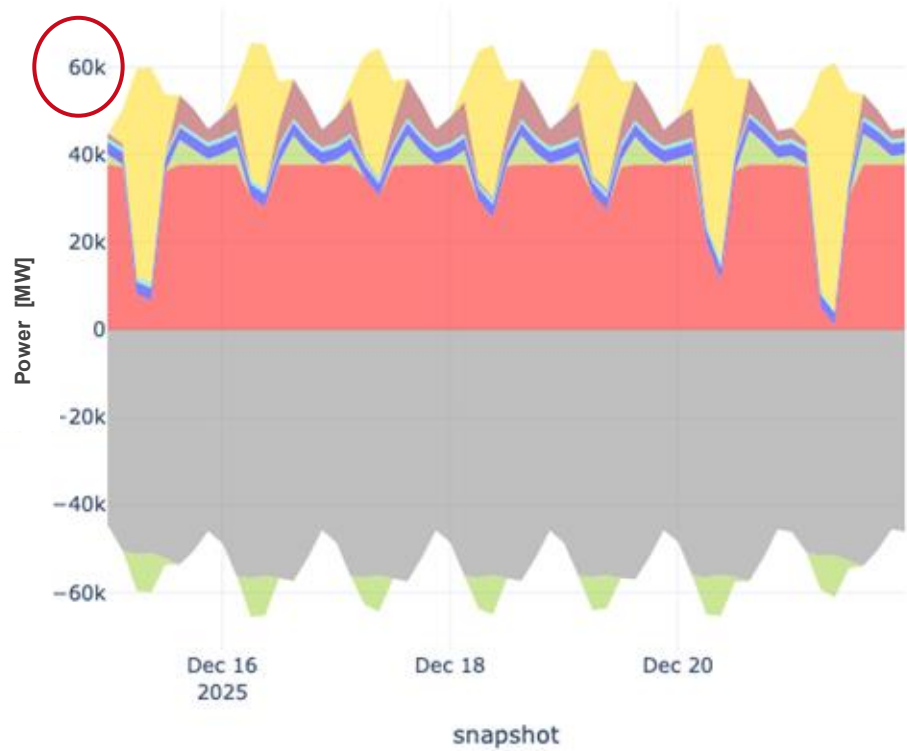


Note: the top 1.5% percent are removed for better visualization (see archive for more detail)

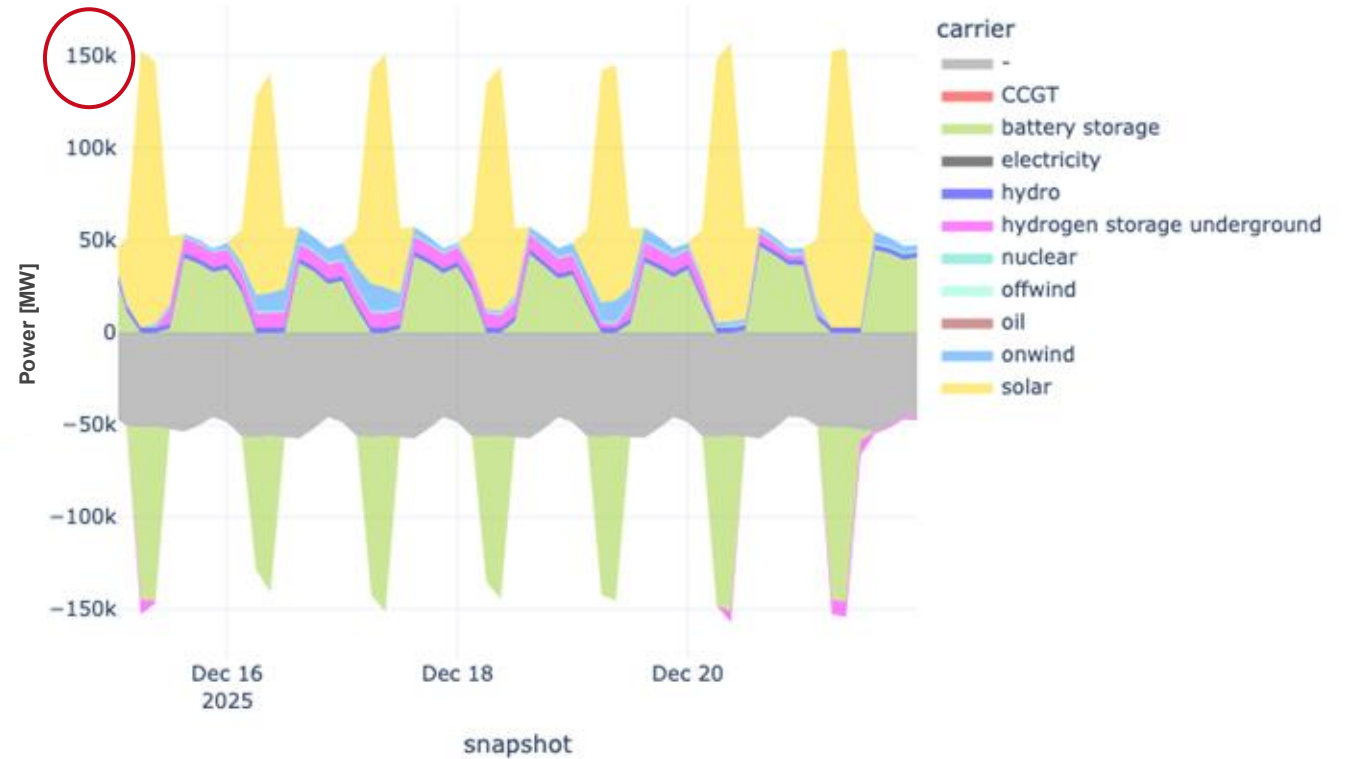
Results - Energy Balance



Energy Balance with CO2



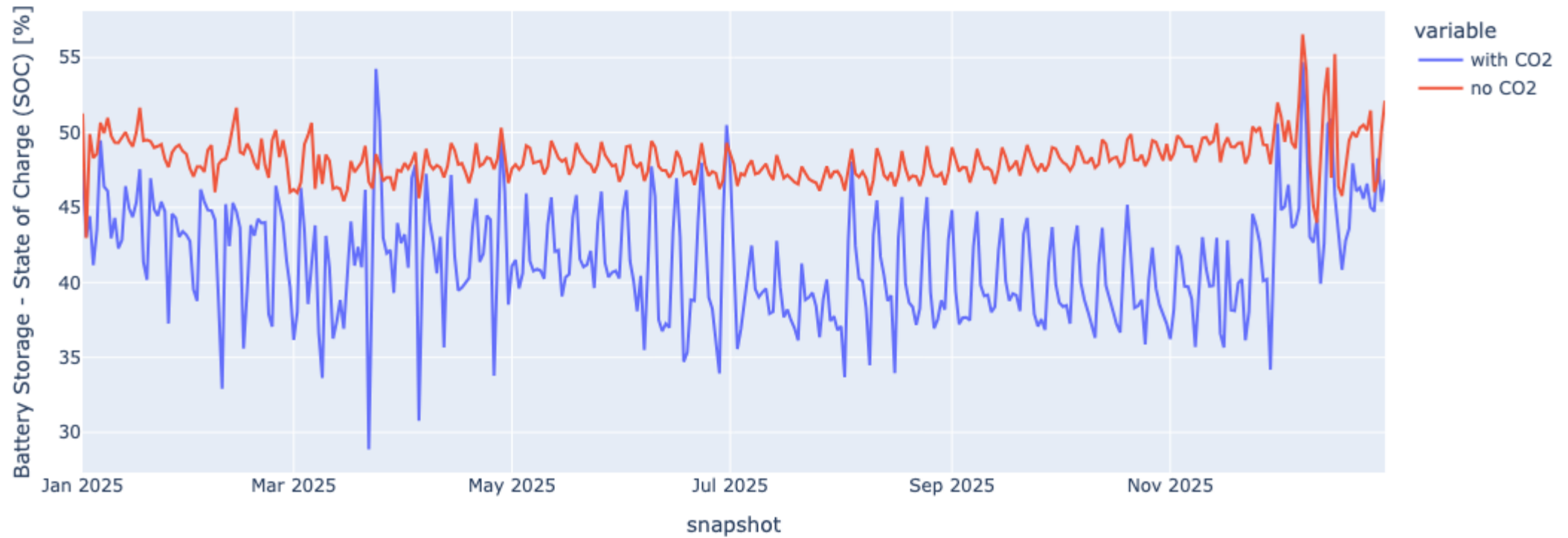
Energy Balance no CO2



Results - Storages



Battery Storage - SOC (Daily Mean)



100% SOC:

55.3 GWh (with CO2)

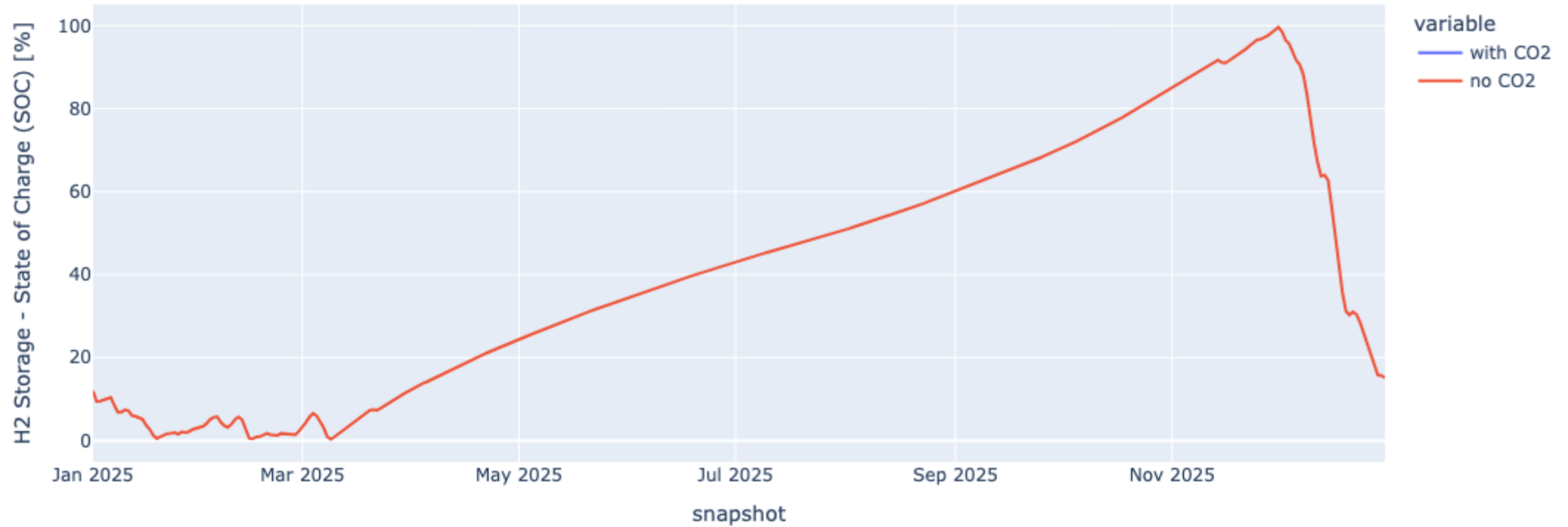
vs.

562.9 GWh (no CO2)

Results - Storages

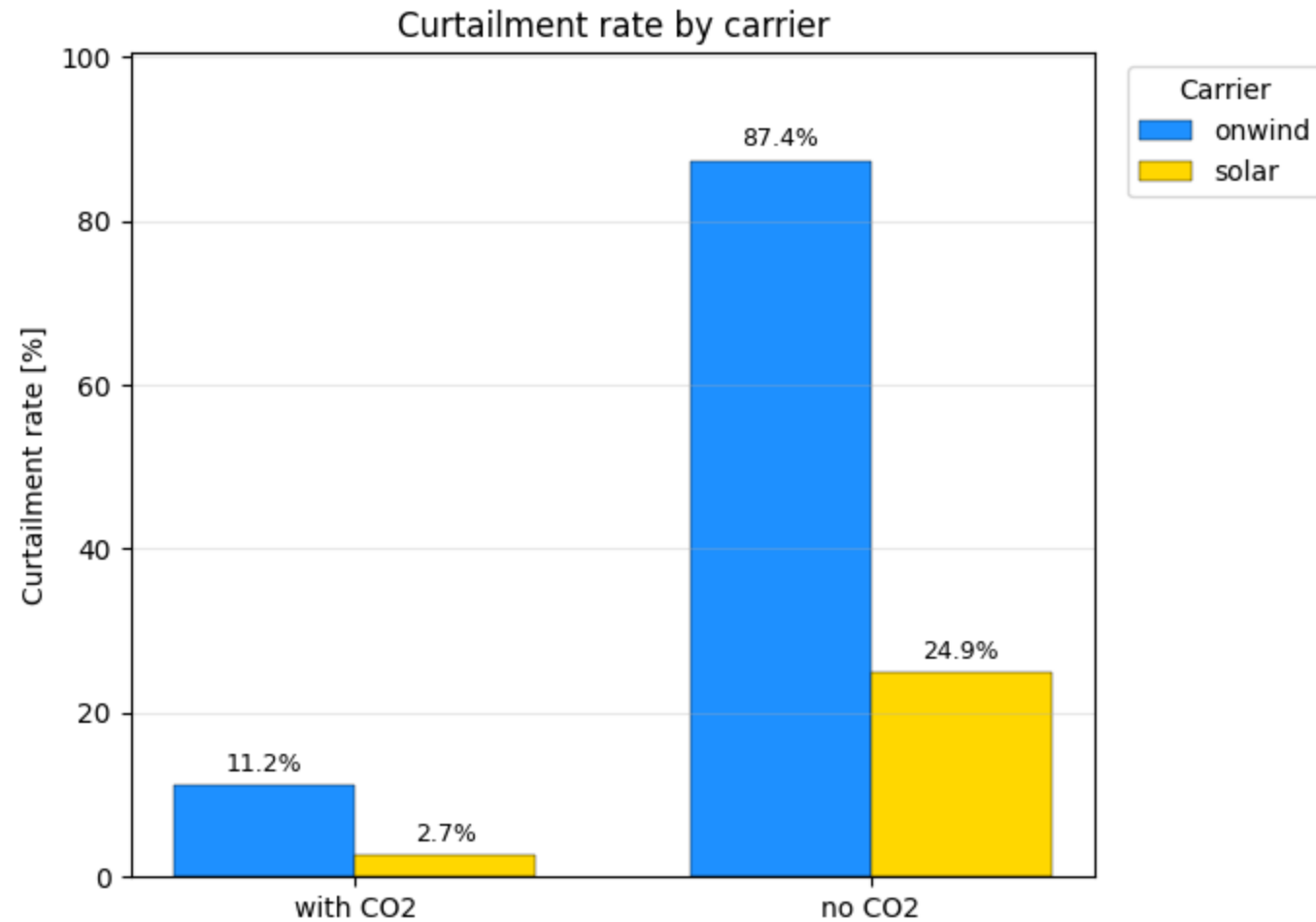


Hydrogen Storage Underground - SOC (Daily Mean)



100% SOC: **5328.6 GWh (no CO2)**

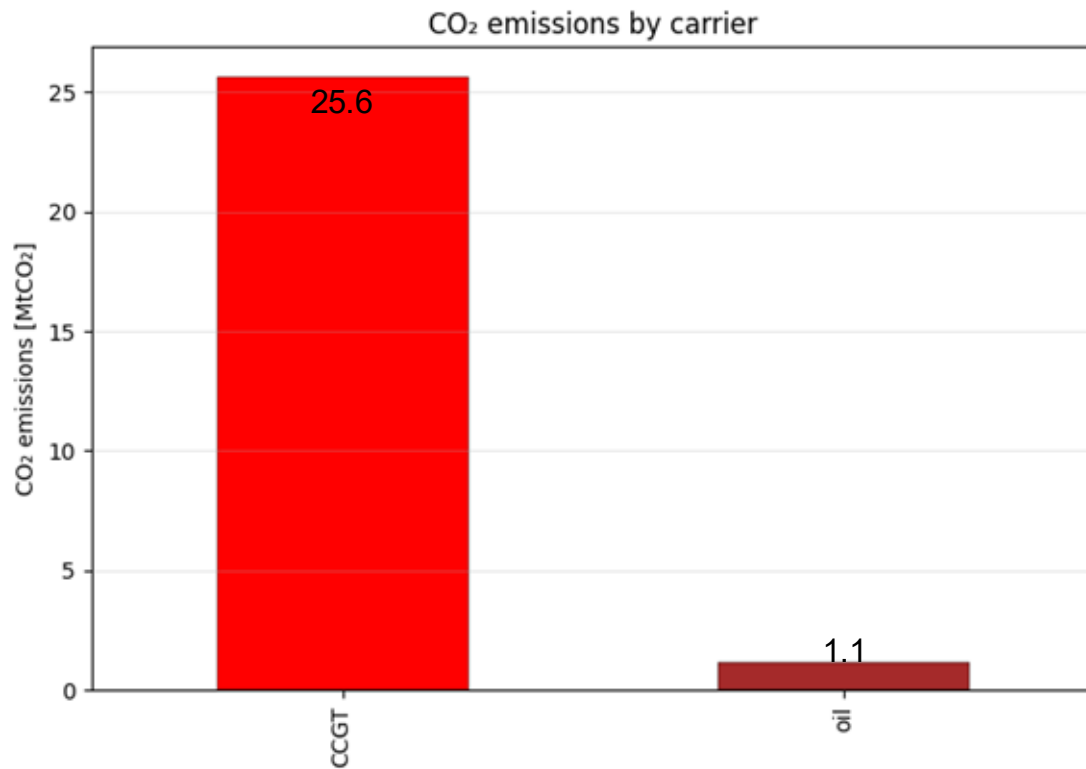
Results - Rate of Curtailment



Results - CO2 Shadow Prices



with CO2 Case:



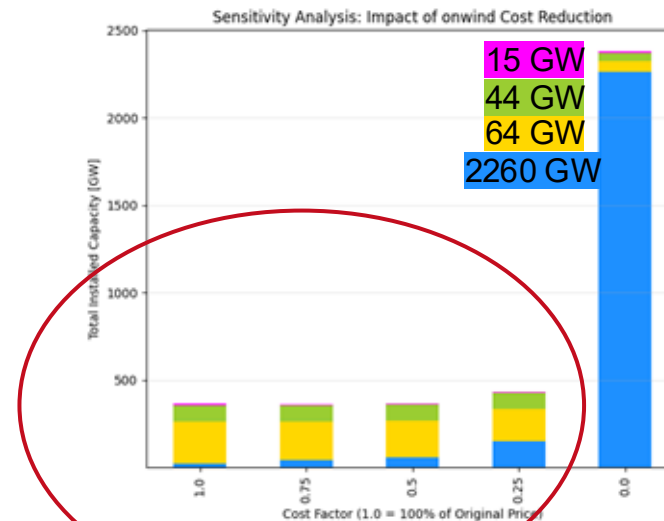
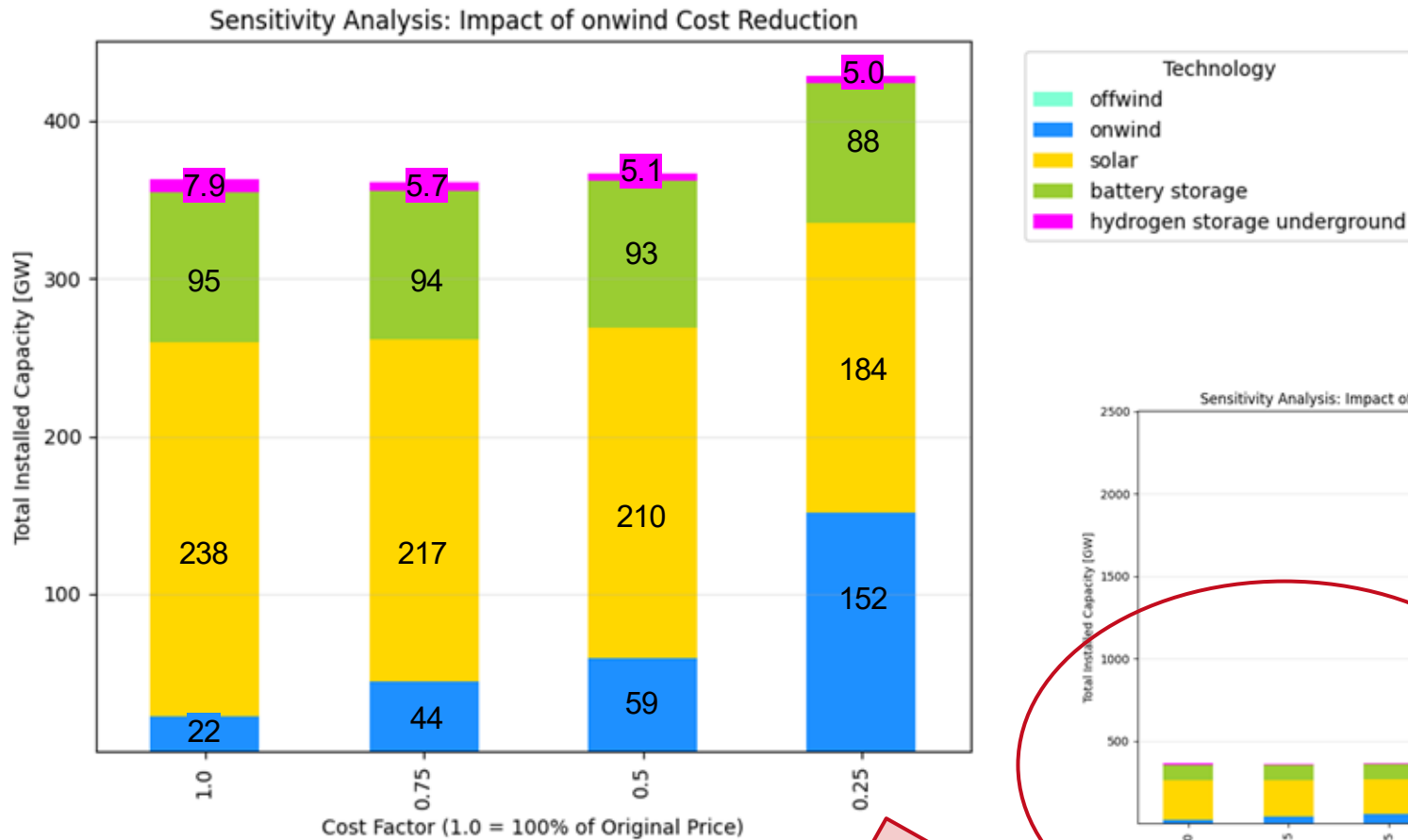
no CO2 Case:

Total emissions: 0.0
tCO₂

Carbon shadow price: 15,389.3
€/tCO₂

→ Allowing 1 extra tonne of CO₂ would reduce
total system cost by ~15,000 €

Sensitivity Analysis - Variation of Onshore Wind Capital Cost



- Lower onwind capex
→ onwind will replace solar & the need for storages become less, because the cheaper onwind makes curtailment more cost efficient than storages

- **Data availability: load data (2013); Power plant data (2017); Population data (2024)**
- **Population distributed load; industry consumption not considered**
- **One bus per region & EEZ**
- Energy import/export not considered
- 2025 costs forecasted from multiple currency years at fixed discount rate
- One cost per technology (sub-type cost difference not considered)
- Storage technologies limited to H2 & BESS
- 3h time resolution
- Dispatchable technology ramping constraints not considered
- **Global fuel prices may overestimate costs for Iran (oil/gas producer)**

Improvement Potential



- Update load data to 2025
- **Allocate regional load by GDP instead of population**
- Update existing power plant data (Global Energy Monitor)
- **Use grid-based network instead of one bus per region**
- Include imports and exports with neighboring countries
- **Further subdivide EEZ regions (beyond Caspian & Persian Gulf)**
- Ensure consistent currency year for costs
- Differentiate gas, oil, and nuclear technologies
- Extend storage technologies beyond batteries and hydrogen tanks
- Increase temporal resolution from 3-hourly to hourly
- **Add technology-specific ramping constraints and costs**
- Consider domestic extraction costs and export opportunity costs

Thanks. Any Question?

Group A | February 5th, 2026

Data Science for Energy System Modelling

External Dataset References

(Bundesamt, 2024)

“Bevölkerung Im Jahr 2024 Um 100 000 Menschen Gewachsen.” *Statistisches Bundesamt*, 2025, www.destatis.de/DE/Presse/Pressemitteilungen/2025/01/PD25_030_124.html. Accessed 02 Feb. 2026.

(Statistical Center of Iran, 2024)

iranopendata.org/en/dataset/iod-06124-estimated-population-iran-province-2024/. Accessed 02 Feb. 2026.

(Ember Energy, 2025)

“Iran.” *Ember*, 10 Apr. 2025, ember-energy.org/countries-and-regions/iran/. Accessed 02 Feb. 2026.

(IR Gas Production, 2024)

Kgi-Admin. “Iran Natural Gas Production: Data and Insights.” *Offshore Technology*, 11 July 2024, www.offshore-technology.com/data-insights/iran-natural-gas-production/. Accessed 02 Feb. 2026.

(IR Oil Production, 2023)

LLC, Globalen. “Iran Percent of World Oil Production - Data, Chart.” *TheGlobalEconomy.Com*, www.theglobaleconomy.com/Iran/oil_production_share/. Accessed 02 Feb. 2026.

The model can be found in git repository: [Link](#)

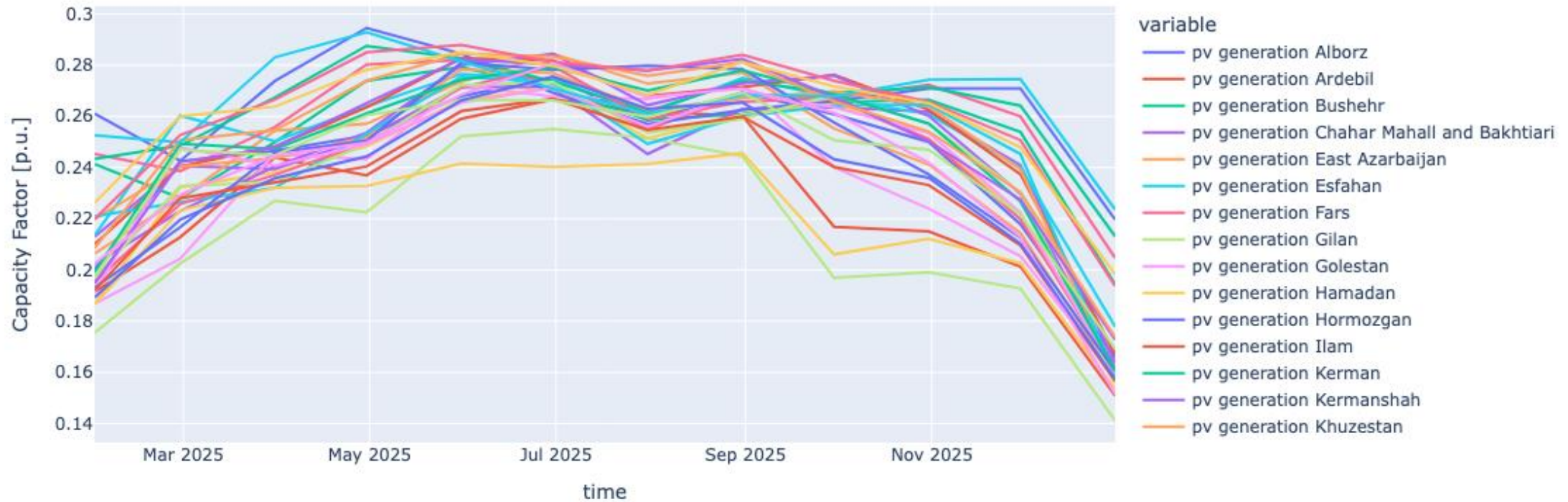
Archive

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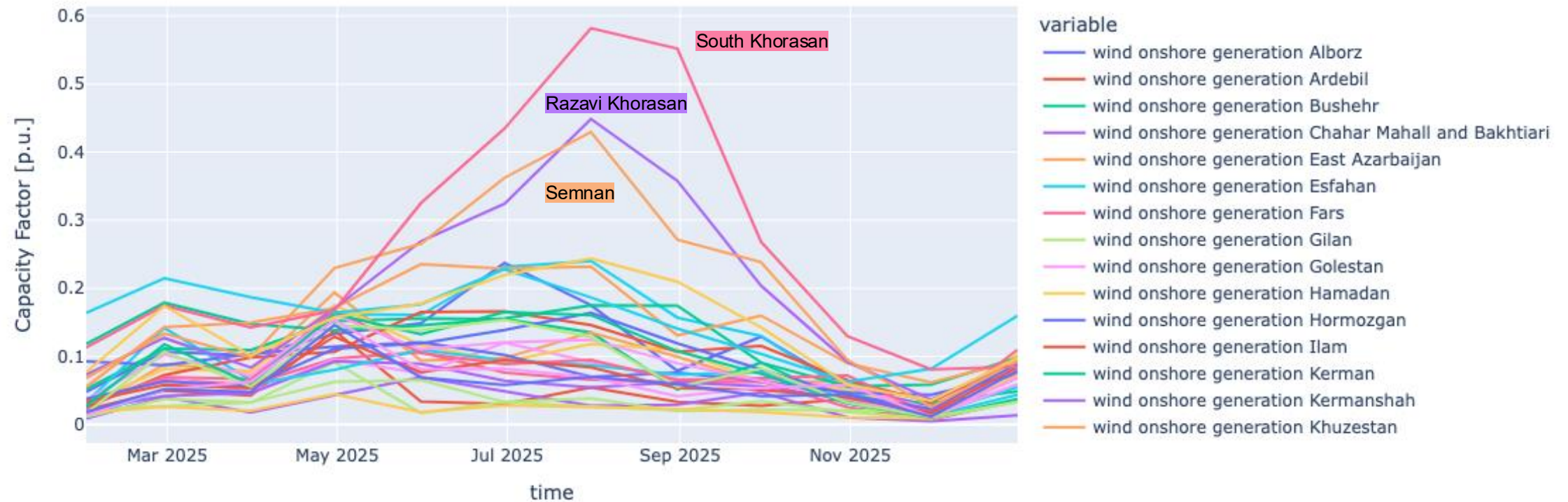
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Data Science for Energy System Modelling

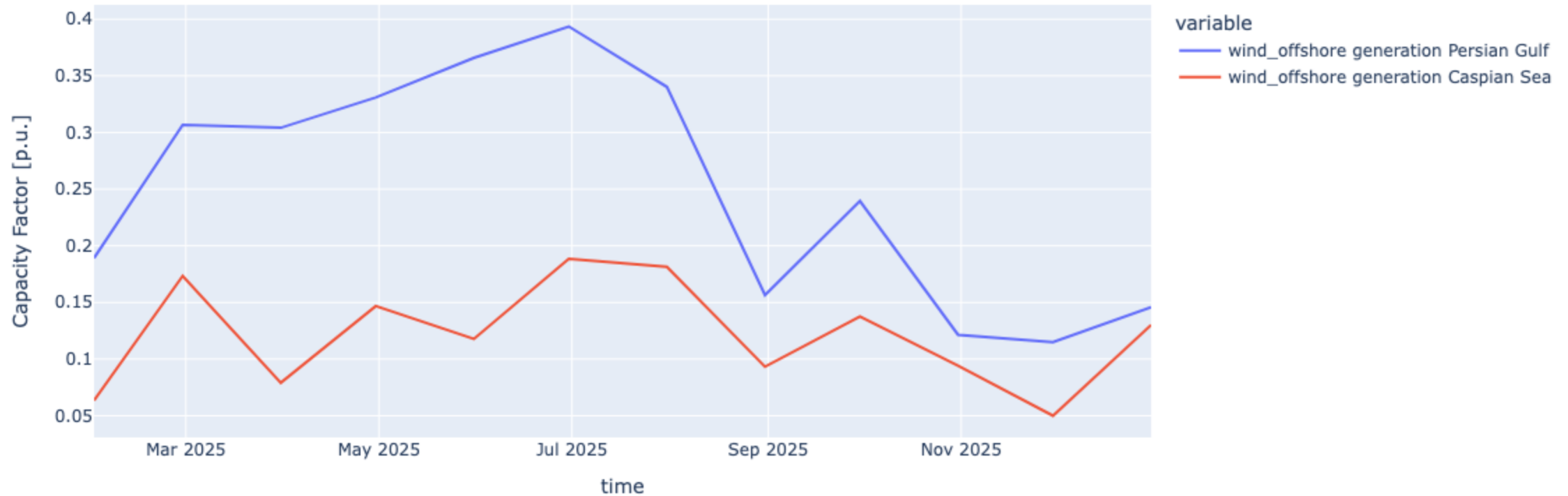
Inputs - RES Time Series (PV - Monthly Average)



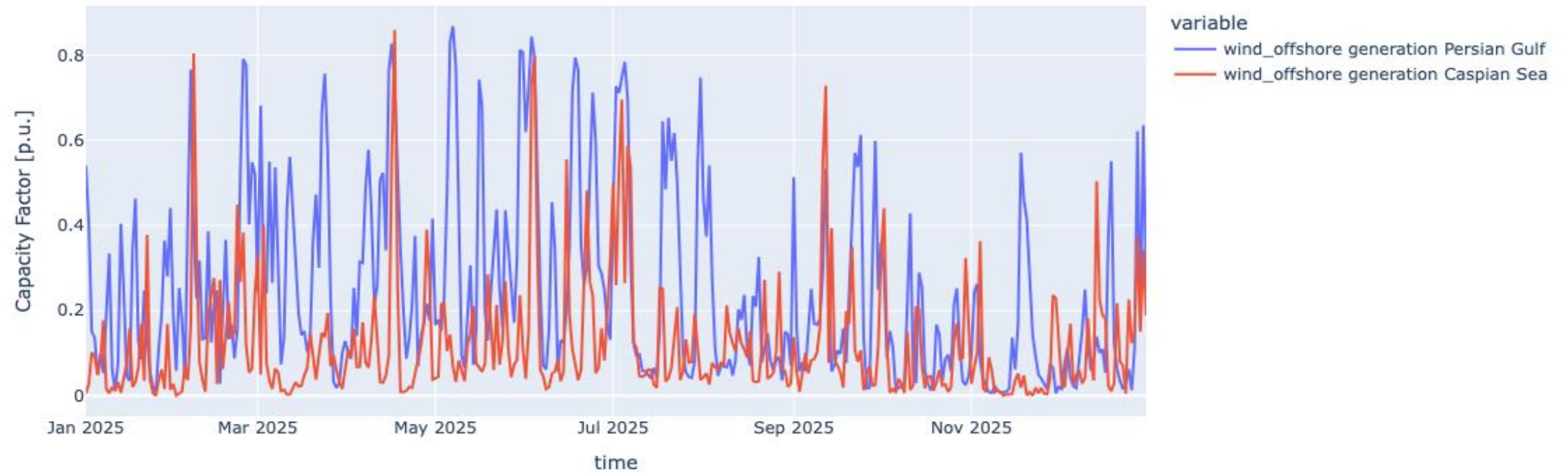
Inputs - RES Time Series (Wind onshore - Monthly Average)



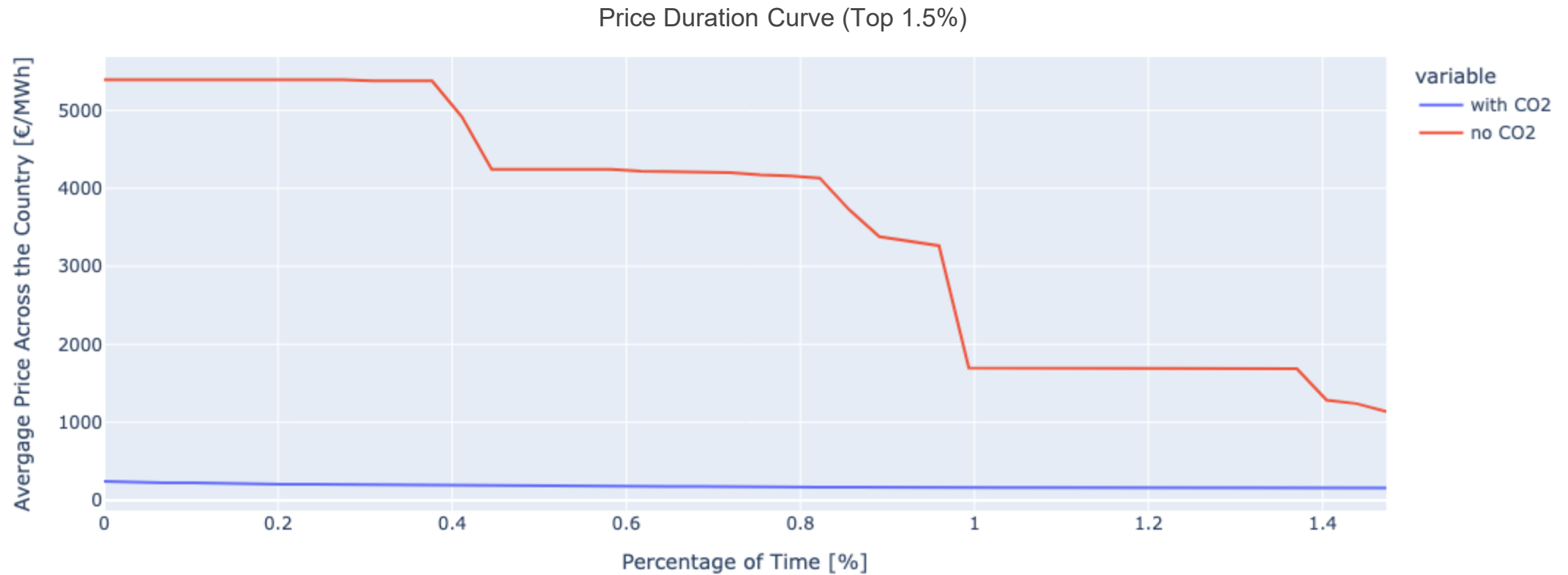
Inputs - RES Time Series (Wind offshore - Monthly Average)



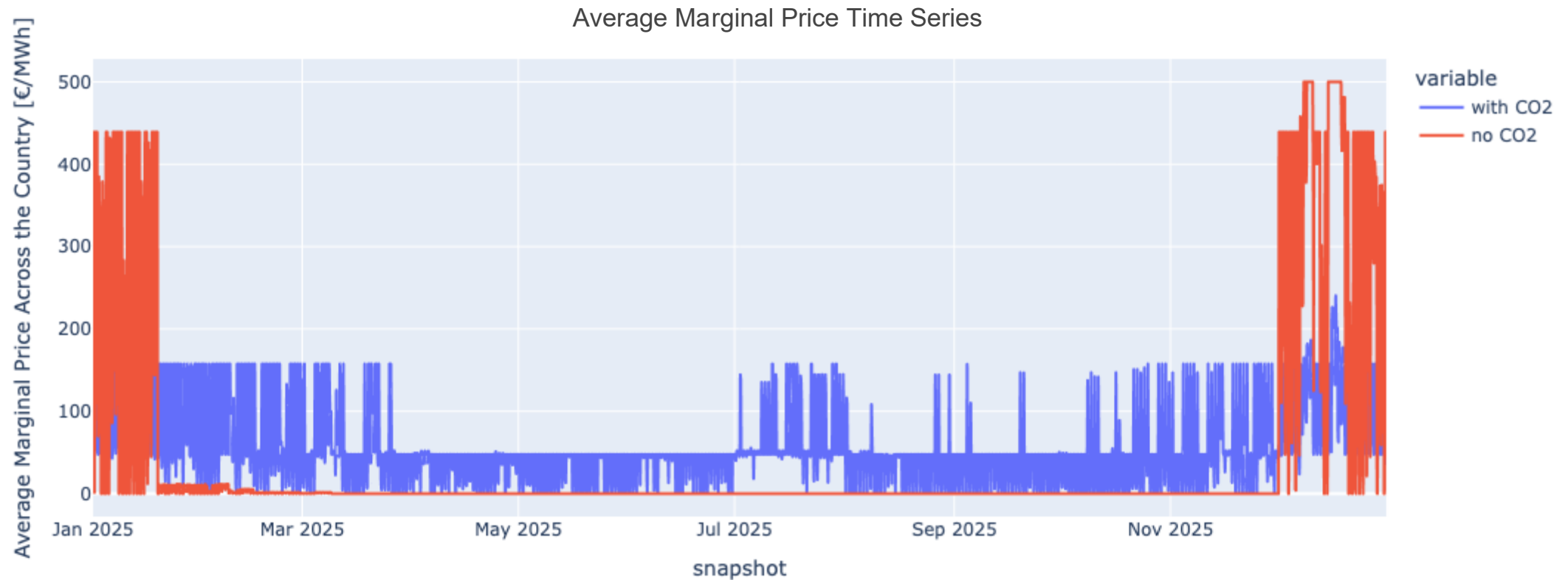
Inputs - RES Time Series (Wind offshore - Daily Average)



Results - Prices

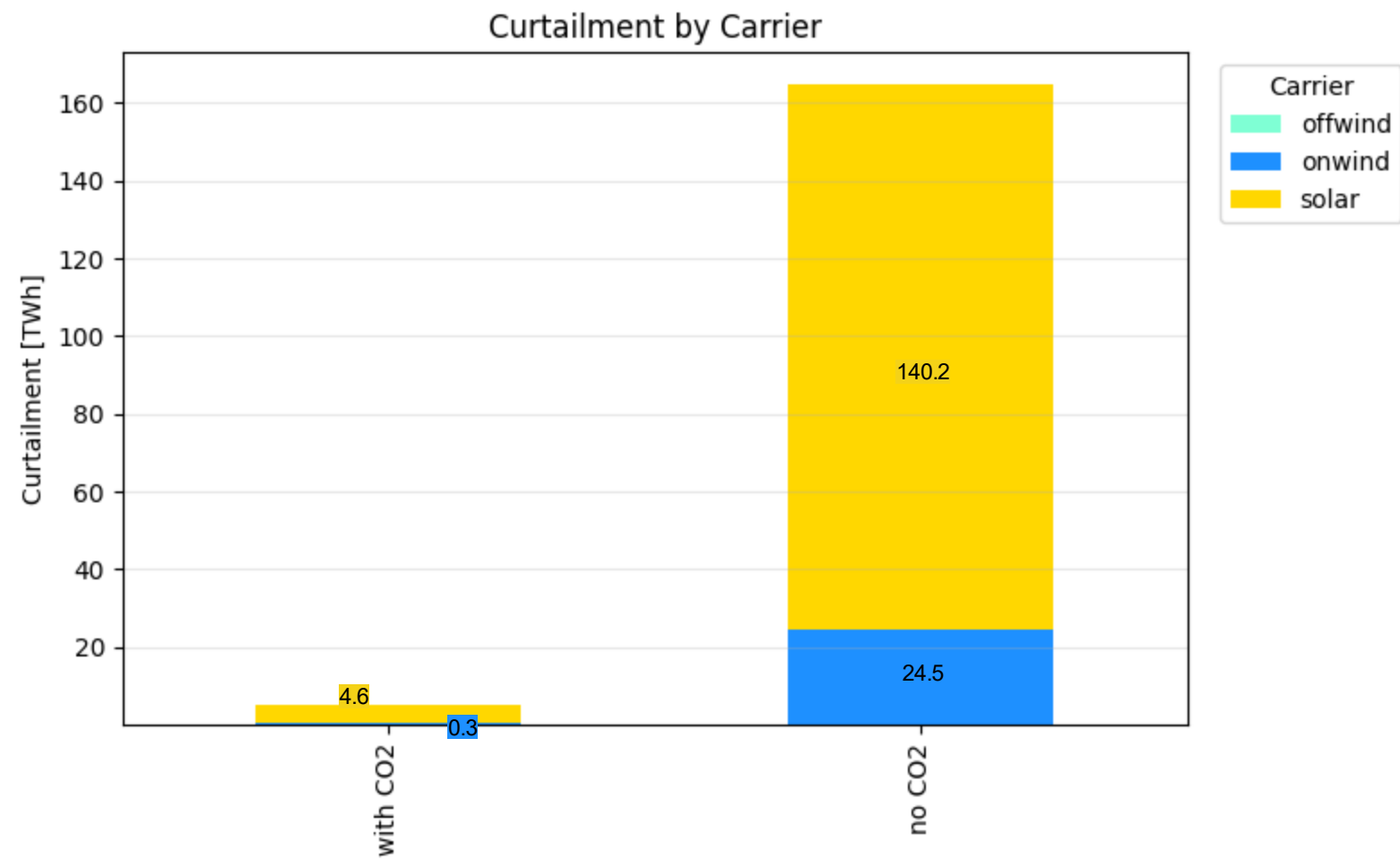


Results - Prices

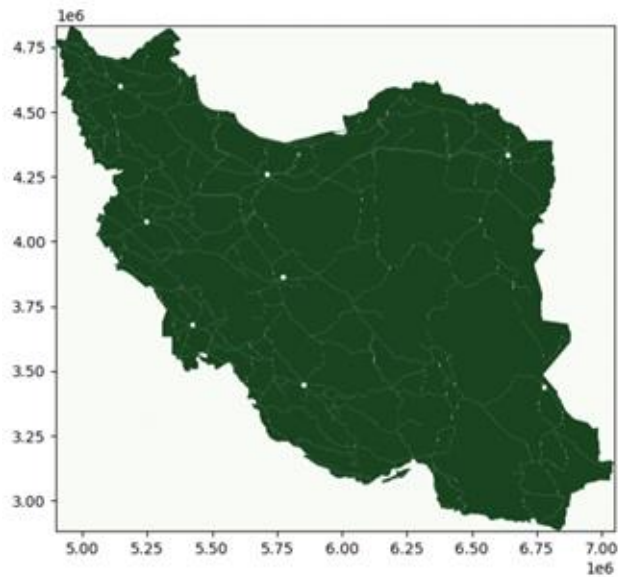


Note: Values greater than 500 €/MWh are clipped at 500/MWh for better visualization

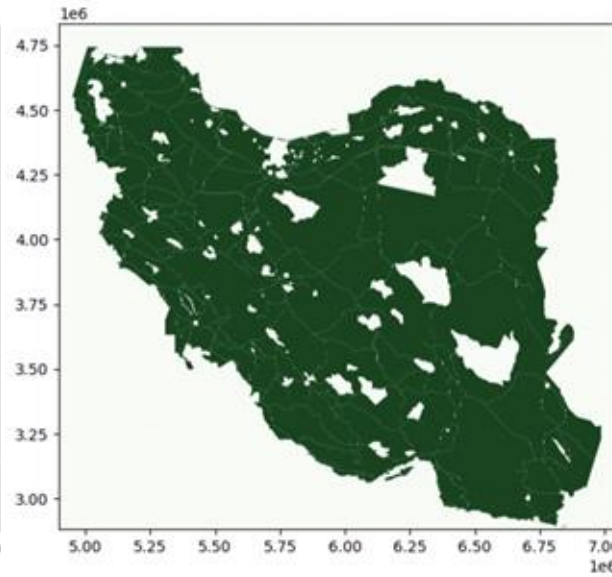
Results - Curtailed Energy



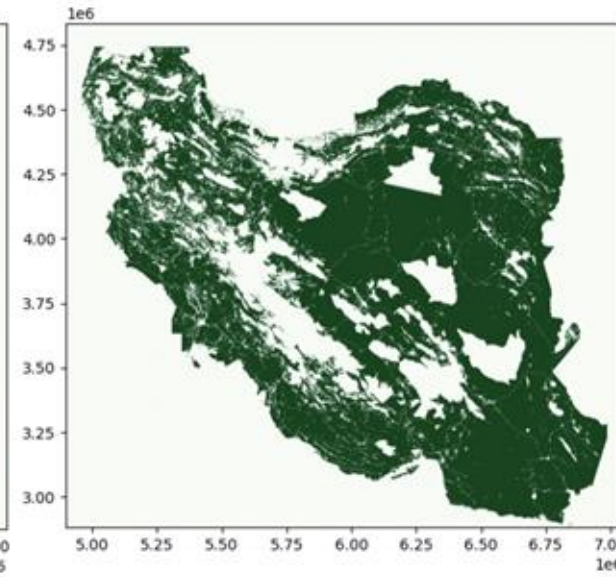
Inputs - Land Eligibility Analysis - Wind Onshore



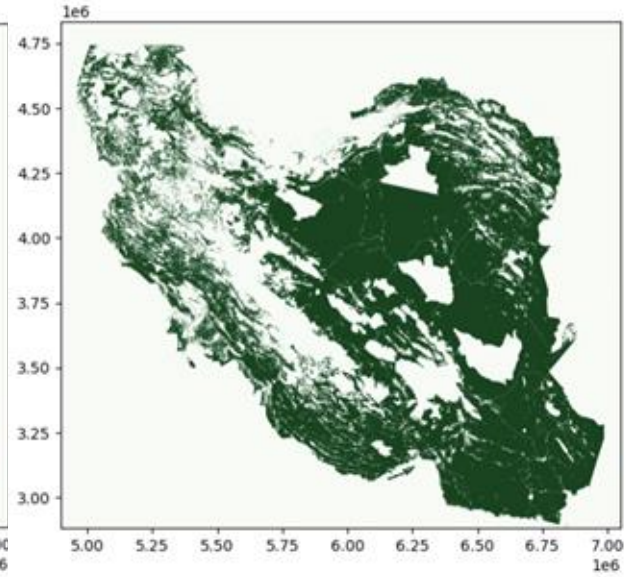
Exclusion of airports (10 km distance)
& Roads (300 m distance)



Exclusion of World protected area



Exclusion of more than 2000 m Elevation
& built up areas (1 km distance)

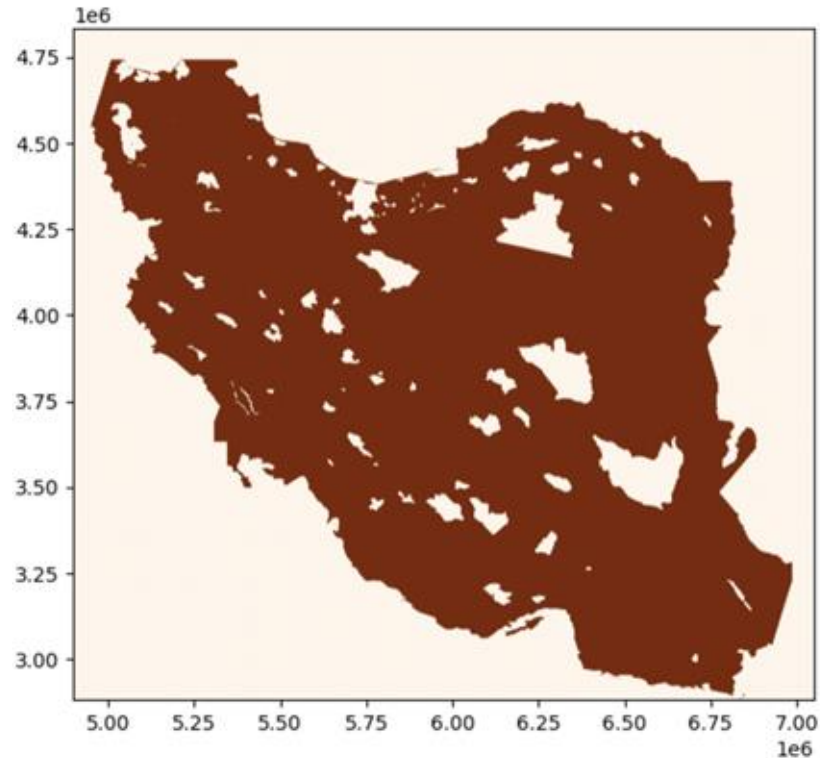


Considering the land classification

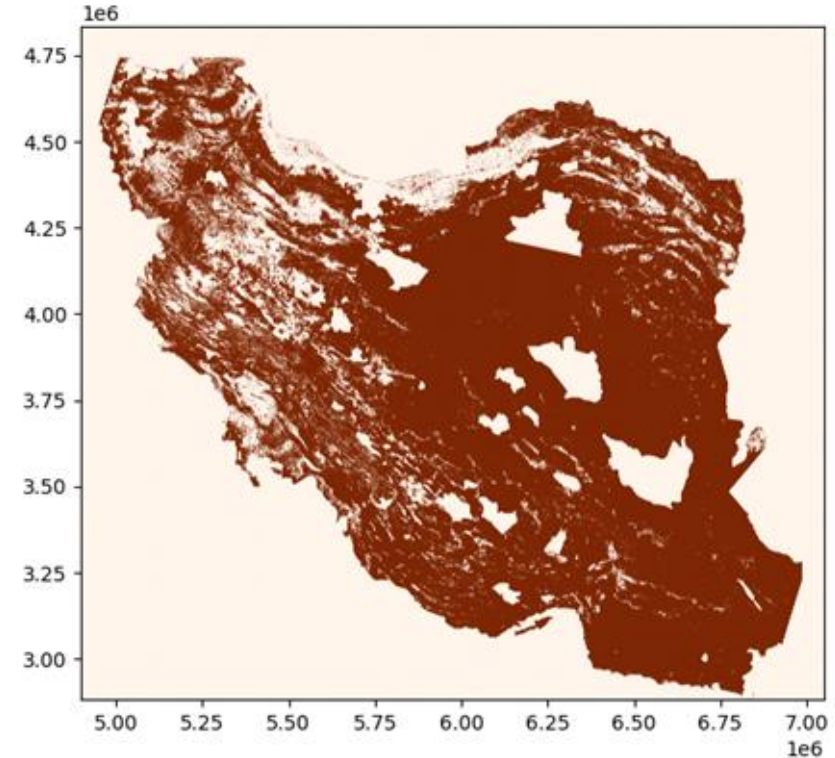
EXCLUDE: codes 100: Moss & Lichen, 90: Wetlands, 200: Open sea, 80: water bodies, 70: snow & ice, 111, 112, 113, 114, 115, 116: closed forest

INCLUDE: codes 60: Bare / sparse vegetation, 20: Shrubs, 30: Herbaceous vegetation

Inputs - Land Eligibility Analysis - PV



Exclusion of World Protected Areas



Considering the Land Classification

EXCLUDE: codes 100: Moss & Lichen, 90: Wetlands, 200: Open sea, 80: water bodies, 70: snow & ice, 111, 112, 113, 114, 115, 116: closed forrest

INCLUDE: rooftop in Built up areas (50), and utility in Shrubs (20), Herbaceous vegetation (30), Bare / sparse vegetation (60)

Inputs - Land Eligibility Analysis - Wind Offshore



Exclusion of World Protected Areas
& distance to shore (10 km)



Water depth (inclusion up to 50 m)